Stochastic and Deterministic Evolution of Moments for a Continuously Monitored Gaussian System

Abstract

Continuously-monitored quantum systems can be leveraged to extract information in a non-destructive manner. In the framework of quantum sensing, this provides an experimentally feasible tool. Although the theory is well developed for systems with Markovian dynamics, a generalization to the non-Markovian setting is not straightforward. In particular, embedding the record of continuous measurement outcomes in the conditional dynamics is dependent on the divisibility of the map, a criterion that is often taken as synonymous with Markovianity. In this work, we develop a framework for the continuous monitoring of a non-Markovian quantum system, namely a bosonic system with Gaussian dynamics. We derive closed equations for the conditional first and second moments of the monitored system. We then build on this framework to design a Bayesian strategy to estimate system and environment parameters such as temperature or external fields.

Contents

1 The general formalism

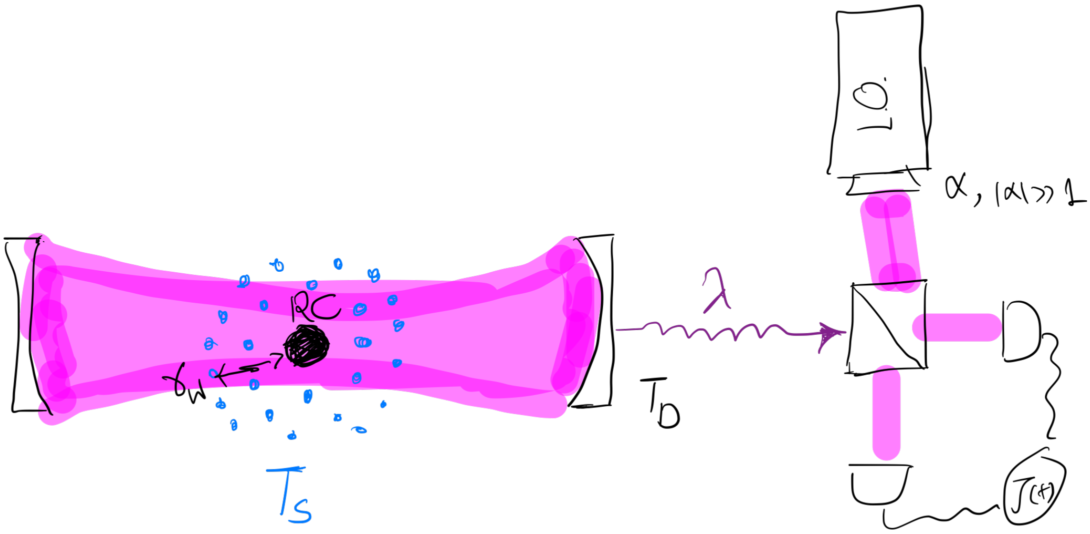


Figure 1: The probe, a cavity mode, is coupled only to a single mode from the sample, the reaction coordinate. In turn, the reaction coordinate is weakly dissipating into a Bosonic sample—a thermal bath at temperature T_S. The cavity is leaking to its surrounding environment, with the dissipation rate λ. There is a detector placed in this environment that is used to register the homodyne current of the leaked signal. This environment is at temperature T_D, which is often take to be zero, but we keep it general. The current $J(t)$ is to be used to learn about sample via a Bayesian update rule.

As depicted in Fig. 1, the main system of interest is a two-mode system, consisting of the cavity mode and the reaction coordinate. The sample, at temperature T_S, is interacting with the reaction coordinate, while the bath where the detector is located only interacts with the cavity mode. We are interested in the dynamics of the reaction coordinate-cavity composit. In particular, how its covariance matrix σ_{12} and displacement vector d_{12} evolve as a function of time, and conditioned on the observed current. Here, 1 and 2 refer to the reaction coordinate and the cavity mode, respectively. If we ignore the information that is gained from the detector, we could describe the dynamics via a standard dissipative master equation with two decay channels. Denote $R^T = [x_1, p_1, x_2, p_2]$, where $\{x_1, p_1\}$ and $\{x_2, p_2\}$ represent the canonical quadratures of the reaction coordinate and probe, respectively. Assume that the RC-probe Hamiltonian reads $H = \frac{1}{2} R^T G R$, with

$$G = \begin{pmatrix} \omega_1^2 & 0 & -g_{12} & 0 \\ 0 & 1 & 0 & 0 \\ -g_{12} & 0 & \omega_2^2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1}$$

The master equation for the first and second moments reads,

$$\begin{aligned} \delta\sigma_{12} &= \delta t(A\sigma_{12} + \sigma_{12}A^T + D), \\ \delta d_{12} &= \delta t(A d_{12}). \end{aligned} \tag{2}$$

where the diffusion and drift matrices are given by

$$A = A_1 + A_2, \quad A_1 = \begin{pmatrix} -\gamma/2 & 1 & 0 & 0 \\ -\omega_1^2 & -\gamma/2 & -g_{12} & 0 \\ 0 & 0 & 0 & 1 \\ -g_{12} & 0 & -\omega_2^2 & 0 \end{pmatrix}, \quad A_2 = 0 \oplus -\frac{\lambda}{2} I,$$

$$D = D_1 + D_2 \quad D_1 = \gamma(2n(T_S, \omega_1) + 1)I \oplus 0, \quad D_2 = 0 \oplus \lambda(2n(T_D, \omega_2) + 1)I, \tag{3}$$

with $n(T, \omega) = [e^{\omega/T} - 1]^{-1}$ being the bosonic occupation number. Here, we have assumed that $\{\lambda\delta t, \gamma\delta t\} \ll 1$.

[Moha]This assumption is absolutely important. Violate it and you will get unphysical results. If you want to take larger dt, γ or λ then use the exact solution to the dynamics instead of this approximation! Specially, the issue is in the subtracted term (information gain) which for large dt (or large lambda) can lead to non-physical cov. mat.

We, however, need to know more than the unconditional state. We need to know the conditional evolution of the composite, as well as the probability of different outcomes (the probability of the observed current) in order to carry on with our Bayesian update. This is not a very complicated matter though. As we just need to keep track of the environment mode that goes into the measurement apparatus and is detected. It is not difficult to see that the dissipation to the detector bath can be fully dilated by bringing in only one mode from the detector environment into the picture. As we show in the appendix, a beam splitter-type interaction between the environment with $\sigma_3 = 2n(T_D, \omega_2) + 1$, $d_3 = 0$, and the mode 2, fully reproduces the dissipative dynamics of the mode 2. We thus have

$$\delta\sigma_{123} = \delta t(A\sigma_{12} + \sigma_{12}A^T + D) \oplus 0 + \lambda\delta t \begin{pmatrix} 0 & 0 & \frac{C}{\sqrt{\lambda\delta t}} \\ 0 & 0 & \frac{\sigma_2 - \sigma_3}{\sqrt{\lambda\delta t}} \\ \frac{C^T}{\sqrt{\lambda\delta t}} & \frac{\sigma_2-\sigma_3}{\sqrt{\lambda\delta t}} & \sigma_2 - \sigma_3 \end{pmatrix},$$

$$\delta d_{123} = \delta t(Ad_{123}) + \sqrt{\lambda\delta t}[0, 0, d_2]^T, \tag{4}$$

where we used the convention

$$\sigma_{123} = \begin{pmatrix} \sigma_1 & C & 0 \\ C^T & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix}, \quad d_{123} = [d_1, d_2, 0]^T. \tag{5}$$

Note that if we trace out the environment mode, we will get back the original equations. Instead, we would like to perform a sharp measurement (Homodyne) on the third mode—which is equivalent to a weak Homodyne current of the second mode—to obtain the current as well as the conditional evolution of the composite. The measurement on the third mode will lead to a feedback on both of the other two modes, as they are all correlated. The feedback on the displacement vector depends on the observed data. The feedback on the covariance matrix only depends on the measurement. In Homodyne measurement (on say, x quadrature), we get the outcome W with probability

$$p(W) \propto \exp\left\{\left(-\frac{(W - \xi)^2}{2V}\right)\right\} = \mathcal{N}(\xi, V), \tag{6}$$

with V being the variance of the measured quadrature, and ξ its displacement. In this case, $\xi = \sqrt{\lambda\delta t}[d_2]_x$, and $V = [\sigma_3]_{xx} + \lambda\delta t([\sigma_2]_{xx} - [\sigma_3]_{xx}) \approx [\sigma_3]_{xx}$. Generally, if we measure using a projective Gaussian measurement, a mode of a system, the remaining modes will be Gaussian. There exist closed relations for what will be their first and second moments based on their initial values, the measurement choice, and the observed outcome. The conditional state of the remaining modes is then [?, ?]

$$\delta\sigma_{12|W} = \delta t[A\sigma_{12} + \sigma_{12}A^T + D] - \frac{\lambda\delta t}{[\sigma_3]_{xx}} \begin{pmatrix} C \\ \sigma_2 - \sigma_3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} C^T & \sigma_2 - \sigma_3 \end{pmatrix}, \tag{7}$$

where the last term is the measurement impact and is independent of the specific outcome. This term particularly reduces the uncertainty in σ_{12} as it incorporates the information gain. Similarly, we have

$$\delta d_{12|W} = \delta t A d_{12} - \frac{\sqrt{\lambda \delta t}}{[\sigma_3]_{xx}} \left(\begin{smallmatrix} C \\ \sigma_2 - \sigma_3 \end{smallmatrix} \right) \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix} \right) (\sqrt{\lambda \delta t} d_2 \ - [W \ \ 0]^T). \tag{8}$$

We can further simplify these to

$$\begin{aligned} \sigma_{12|W} &= \sigma_{12} + \delta t[A\sigma_{12} + \sigma_{12}A^T + D] - \lambda \delta t K K^T \\ d_{12|W} &= d_{12} + \delta t A d_{12} + (W - \sqrt{\lambda \delta t}[d_2]_x)\sqrt{\lambda \delta t} K, \\ K &= \frac{1}{[\sigma_3]_{xx}} \left(\begin{smallmatrix} C \\ \sigma_2 - \sigma_3 \end{smallmatrix} \right) \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right). \end{aligned} \tag{9}$$

Note that we can define $\delta Z = \sqrt{\delta t}(W - [d_2]_x \sqrt{\lambda \delta t})$. This is a Wiener process, as its expectation value vanishes, but its variance is $E(\delta Z^2) = \delta t[\sigma_3]_{xx}$. As such, by substitution for δZ, we have

$$d_{12|\delta Z} = d_{12} + \delta t A d_{12} + \sqrt{\lambda} \delta Z K. \tag{10}$$

Remark.—Note that $W(t)$ at each time do not have independent Gaussian distributions, as their mean depends on the previous outcome. This is not the case for δZ as we discussed above.

2 The HD current

We can define the current as follows. We first define a new variable $J = W/\sqrt{\lambda \delta t}$. This variable can be written as its fluctuations around its mean value $\langle J(t) \rangle = [d_2(t)]_x$,

$$J(t) = \frac{W(t)}{\sqrt{\lambda \delta t}} = [d_2(t)]_x + \frac{\delta Z}{\sqrt{\lambda \delta t}}. \tag{11}$$

In this regard, we really don't need to work with δZ, as it can be seen, the current is nothing but the outcome history $W(t)$ up to a constant.

2.1 General Gaussian measurement

If instead of Homodyne, we were performing a Gaussian measurement with Cov. Mat. σ^M, then the post measurement state would be different. But also, the outcome probability will be different too. In this case, we would have $p(W) = \mathcal{N}(\xi, V)$, with $\xi = \sqrt{\lambda \delta t} d_2$, and $V = \sigma_3 + \lambda \delta t(\sigma_2 - \sigma_3) + \sigma^M$, which is a matrix, and its moore-penrose inverse is needed to ontain the multi-variate normal distribution. The post measurement cov. mat. and dis. vec. of the other modes read

$$\begin{aligned} \delta\sigma_{12|W} &= \delta t[A\sigma_{12} + \sigma_{12}A^T + D] - \lambda \delta t \left(\begin{smallmatrix} C \\ \sigma_2 - \sigma_3 \end{smallmatrix} \right) V^{-1} \left(\begin{smallmatrix} C^T & \sigma_2 - \sigma_3 \end{smallmatrix} \right) \\ \delta d_{12|W} &= \delta t A d_{12} + \sqrt{\lambda \delta t}(W - \sqrt{\lambda \delta t} d_2) V^{-1} \left(\begin{smallmatrix} C^T & \sigma_2 - \sigma_3 \end{smallmatrix} \right) \end{aligned} \tag{12}$$

3 Equivalence of a Lindbladian Master equation with a beam splitter interaction

A single mode (extension is trivial) can undergo an open quantum system dynamics with Lindbladian master equation

$$\begin{aligned} \sigma &\mapsto e^{-\gamma t}\sigma + (1 - e^{-\gamma t})\sigma_3, \\ d &\mapsto e^{-\gamma t/2} d, \end{aligned} \tag{13}$$

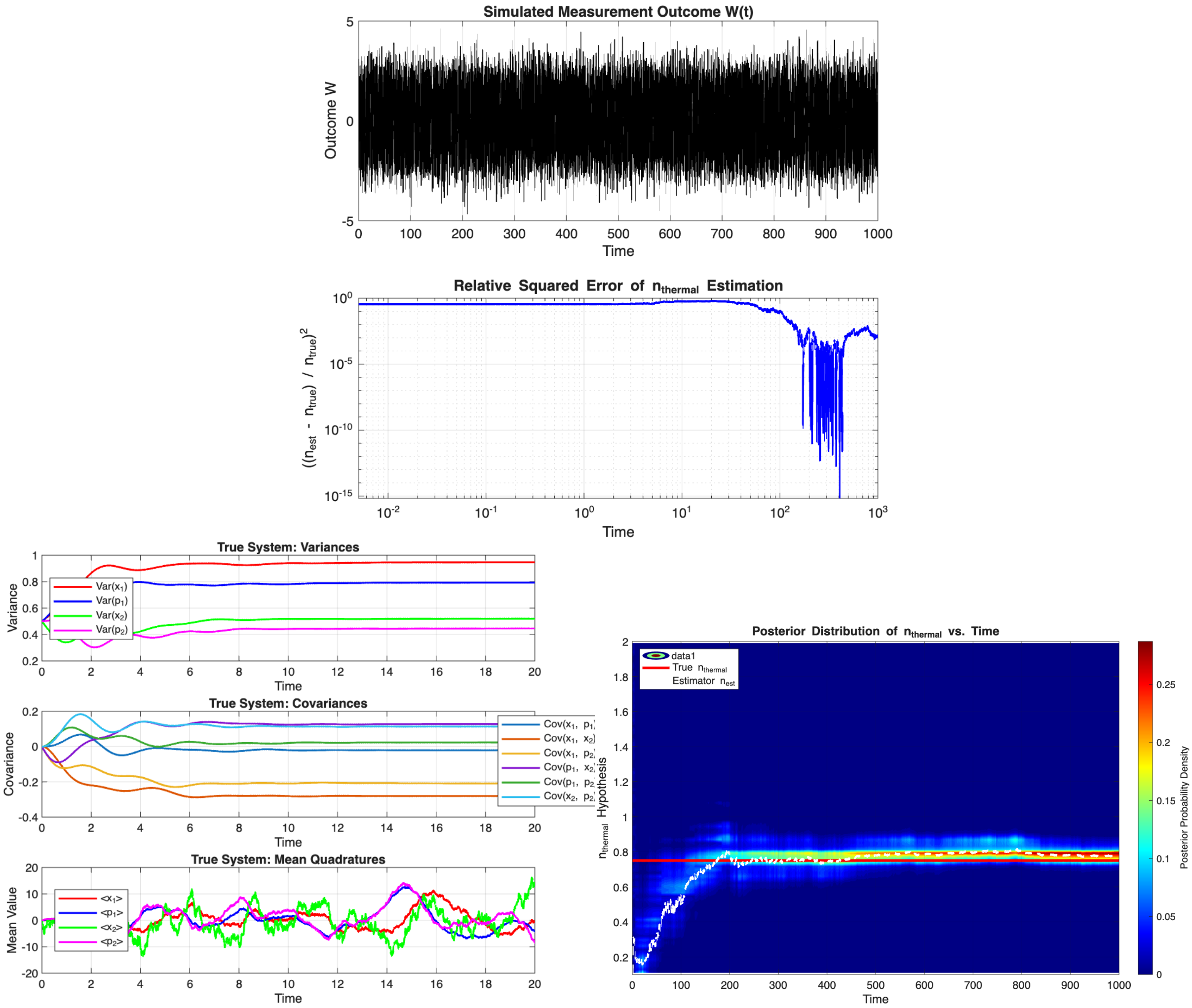


Figure 2: Enter Caption

where σ_3 is the steady state of the system. If the environment is thermal then this will be $\sigma_3 = \nu I$, with $\nu = \coth(\omega/2T)$. The term γ is the dissipation rate $\gamma = J(\omega)$, where $J(\omega)$ is the spectral density at the system's frequency ω.

This dynamics is reproducible by noting that it is resembling a partial swap between the system and the steady state. One can simply overlap the system and σ_3 on a Beam splitter with transmissivity $\tau = e^{-\gamma t}$ to get

$$\begin{aligned}
\left[\begin{smallmatrix}\sigma & 0\\ 0 & \sigma_3\end{smallmatrix}\right] &\mapsto S_{BS}(e^{-\gamma t})\left[\begin{smallmatrix}\sigma & 0\\ 0 & \sigma_3\end{smallmatrix}\right] S_{BS}^T(e^{-\gamma t})\\
&= \left[\begin{smallmatrix} e^{-\gamma t}\sigma+(1-e^{-\gamma t})\sigma_3 & \sqrt{e^{-\gamma t}(1-e^{-\gamma t})}(\sigma-\sigma_3)\\ \sqrt{e^{-\gamma t}(1-e^{-\gamma t})}(\sigma-\sigma_3) & e^{-\gamma t}\sigma_3+(1-e^{-\gamma t})\sigma\end{smallmatrix}\right],\\
\left[\begin{smallmatrix} d\\ 0\end{smallmatrix}\right] &\mapsto S_{BS}(e^{-\gamma t})\left[\begin{smallmatrix} d\\ 0\end{smallmatrix}\right] = \left[\begin{smallmatrix} e^{-\gamma t/2}d\\ \sqrt{1-e^{-\gamma t/2}}d\end{smallmatrix}\right].
\end{aligned} \tag{14}$$

Clearly, by tracing out the environment mode we will get back the correct master equation for the system. However, we would like to keep the environment mode, as we will exactly perform measurements on this environment mode and then update the information to find a conditional state of the system.

Note that, at the level of infinitesimal evolution, which is what we need exactly for continious monitoring, the above transformations to the leading order reads

$$\begin{aligned}
\left[\begin{smallmatrix}\sigma & 0\\ 0 & \sigma_3\end{smallmatrix}\right] &\mapsto \left[\begin{smallmatrix}(1-\gamma dt)\sigma+\gamma dt\sigma_3 & \sqrt{\gamma dt}(\sigma-\sigma_3)\\ \sqrt{\gamma dt}(\sigma-\sigma_3) & (1-\gamma dt)\sigma_3+\gamma dt\sigma\end{smallmatrix}\right],\\
\left[\begin{smallmatrix} d\\ 0\end{smallmatrix}\right] &\mapsto \left[\begin{smallmatrix}(1-\gamma dt/2)d\\ \sqrt{\gamma dt/2d}\end{smallmatrix}\right].
\end{aligned} \tag{15}$$

If σ itself was a two mode system $\sigma_{12} = \left(\begin{smallmatrix}\sigma_1 & C\\ C^T & \sigma_2\end{smallmatrix}\right)$, where only the second mode undergoes the dissipation, then

$$\begin{aligned}
\left[\begin{smallmatrix}\sigma_{12} & 0\\ 0 & \sigma_3\end{smallmatrix}\right] &\mapsto I\oplus S_{BS}(e^{-\gamma t})\left[\begin{smallmatrix}\sigma_{12} & 0\\ 0 & \sigma_3\end{smallmatrix}\right] I\oplus S_{BS}^T(e^{-\gamma t})\\
&= \left[\begin{smallmatrix}\sigma_1 & \sqrt{e^{-\gamma dt}}C & \sqrt{1-e^{-\gamma dt}}C\\ \sqrt{e^{-\gamma dt}}C^T & e^{-\gamma dt}\sigma_2+(1-e^{-\gamma dt})\sigma_3 & \sqrt{e^{-\gamma dt}(1-e^{-\gamma dt})}(\sigma_2-\sigma_3)\\ \sqrt{1-e^{-\gamma dt}}C^T & \sqrt{e^{-\gamma dt}(1-e^{-\gamma dt})}(\sigma_2-\sigma_3) & e^{-\gamma dt}\sigma_3+(1-e^{-\gamma dt})\sigma_2\end{smallmatrix}\right],\\
\left[\begin{smallmatrix} d_{12}\\ 0\end{smallmatrix}\right] &\mapsto I\oplus S_{BS}(e^{-\gamma t})\left[\begin{smallmatrix} d_{12}\\ 0\end{smallmatrix}\right] = \left[\begin{smallmatrix} d_1\\ e^{-\gamma t/2}d_2\\ \sqrt{1-e^{-\gamma t}}d_2\end{smallmatrix}\right].
\end{aligned} \tag{16}$$

For small $\gamma\delta t \ll 1$, we can expand to get

$$\begin{aligned}
\delta\sigma_{123} &= \lambda\delta t\begin{pmatrix} 0 & -\frac{C}{2} & \frac{C}{\sqrt{\lambda\delta t}}\\ -\frac{C^T}{2} & \sigma_3-\sigma_2 & \frac{\sigma_2-\sigma_3}{\sqrt{\lambda\delta t}}\\ \frac{C^T}{\sqrt{\lambda\delta t}} & \frac{\sigma_2-\sigma_3}{\sqrt{\lambda\delta t}} & \sigma_2-\sigma_3\end{pmatrix}\\
&= \delta t\left[A_2\oplus 0\sigma_{123}+\sigma_{123}A_2\oplus 0+D_2\oplus 0\right]+\lambda\delta t\begin{pmatrix} 0 & 0 & \frac{C}{\sqrt{\lambda\delta t}}\\ 0 & 0 & \frac{\sigma_2-\sigma_3}{\sqrt{\lambda\delta t}}\\ \frac{C^T}{\sqrt{\lambda\delta t}} & \frac{\sigma_2-\sigma_3}{\sqrt{\lambda\delta t}} & \sigma_2-\sigma_3\end{pmatrix}\\
d\delta d_{123} &= \lambda\delta t\begin{bmatrix} 0\\ -\frac{d_2}{2}\\ \frac{d_2}{\sqrt{\lambda\delta t}}\end{bmatrix} = \delta t(A_2\oplus 0d_{123})+\sqrt{\lambda\delta t}[0,0,d_2]^T,
\end{aligned} \tag{17}$$

where we defined $A_2 = 0\oplus(-\lambda/2)I$ and $D_2 = 0\oplus\lambda\sigma_3$.

4 The Hamiltonian

We will model the interaction between the system and the bath with a Hamiltonian of the form

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB}, \tag{18}$$

where

$$\hat{H}_S = \frac{1}{2}(p_S^2 + \Omega_S^2 x_S^2) + \frac{1}{2}\sum_k \frac{c_k^2}{\omega_k^2} s^2, \tag{19}$$

$$\hat{H}_B = \frac{1}{2}\sum_k (p_k^2 + \omega_k^2 x_k^2), \tag{20}$$

$$\hat{H}_{SB} = -x_S \sum_k c_k x_k, \tag{21}$$

such that it can be rewritten in Brownian motion form

$$\hat{H} = \frac{1}{2}(p_S^2 + \Omega_S^2 x_S^2) + \frac{1}{2}\sum_k \left[p_k^2 + \omega_k^2 \left(x_k - \frac{c_k}{\omega_k^2} x_S \right)^2 \right]. \tag{22}$$

Let us now study 3 different approaches to solve the dynamics of the system.

5 Approach 1: the exact solution and the Langevin equation

In this section, we solve exactly the equations of motion (i.e. Langevin equations) of the system quadratures. One can show that the equations of motion for the system and bath modes read

$$\ddot{x}_{\mathrm{S}}(t) = -\Omega_{\mathrm{S}}^2 x_{\mathrm{S}}(t) - x_{\mathrm{S}}(t) \sum_k \frac{c_k^2}{\omega_k^2} + \sum_k x_k(t) c_k, \tag{23}$$

$$\ddot{x}_k(t) = -\omega_k^2 x_k(t) + c_k x_{\mathrm{S}}(t). \tag{24}$$

For that purpose, let us first solve Eq. (24) using standard differential equation techniques. First, we know that the set of solutions of the homogeneous equation is the one for the harmonic oscillator, i.e. $\{x_k^{(1)}(t), x_k^{(2)}(t)\} = \{\sin(\omega_k t), \cos(\omega_k t)\}$. The particular solution will be given by

$$x_k^p(t) = u_k^{(1)}(t) x_k^{(1)}(t) + u_k^{(2)}(t) x_k^{(2)}(t), \tag{25}$$

where $u_k^{(1)}(t)$ and $u_k^{(2)}(t)$ are defined (in terms of their derivatives w.r.t. t, that we denote with a dot) as follows:

$$\dot{u}_k^{(1)} = \frac{\begin{vmatrix} 0 & x_k^{(2)}(t) \\ c_k x_{\mathrm{S}}(t) & \dot{x}_k^{(2)}(t) \end{vmatrix}}{W[x_k^{(1)}(t), x_k^{(2)}(t)]}, \quad \dot{u}_k^{(2)} = \frac{\begin{vmatrix} x_k^{(1)}(t) & 0 \\ \dot{x}_k^{(1)}(t) & c_k x_{\mathrm{S}}(t) \end{vmatrix}}{W[x_k^{(1)}(t), x_k^{(2)}(t)]}, \tag{26}$$

with $W[x_k^{(1)}(t), x_k^{(2)}(t)]$ being the determinant of the Wronskian matrix. To obtain $u_k^{(1)}(t)$ and $u_k^{(2)}(t)$ we just need to integrate the previous equations.

For our problem, we can show them to be

$$u_k^{(1)}(t) = -\frac{c_k}{\omega_k} \int_0^t \mathrm{d}t'\, x_{\mathrm{S}}(t') \sin\big(\omega_k t'\big), \tag{27}$$

$$u_k^{(2)}(t) = \frac{c_k}{\omega_k} \int_0^t \mathrm{d}t'\, x_{\mathrm{S}}(t') \cos\big(\omega_k t'\big). \tag{28}$$

Therefore, the final solution of Eq. (24) (including the initial conditions) is given by

$$x_k(t) = x_k(0)\cos(\omega_k t) + \frac{p_k(0)}{\omega_k}\sin(\omega_k t) + \frac{c_k}{\omega_k}\int_0^t \mathrm{d}t'\, x_\mathrm{S}(t')\sin\left[\omega_k(t-t')\right], \tag{29}$$

where we have used that $\sin(A-B) = \sin A\cos B - \sin B\cos A$. Now, by plugging Eq. (29) into Eq. (23) we get:

$$\begin{aligned}\ddot{x}_\mathrm{S}(t) + \Omega_\mathrm{S}^2 x_\mathrm{S}(t) + x_\mathrm{S}(t)\sum_k \frac{c_k^2}{\omega_k^2} &= \sum_k c_k\left(x_k(0)\cos(\omega_k t) + \frac{p_k(0)}{\omega_k}\sin(\omega_k t)\right) \\ &+ \sum_k \frac{c_k^2}{\omega_k}\int_0^t \mathrm{d}t'\, x_\mathrm{S}(t')\sin\left[\omega_k(t-t')\right].\end{aligned} \tag{30}$$

By defining the bath spectral density

$$\mathcal{J}(\omega) = \frac{\pi}{2}\sum_k \frac{c_k^2}{\omega_k}\delta(\omega-\omega_k), \tag{31}$$

and rearranging terms and we get

$$\ddot{x}_\mathrm{S}(t) + (\Omega_\mathrm{S}^2 + \delta\Omega_\mathrm{S}^2)x_\mathrm{S}(t) - \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\,\mathcal{J}(\omega)\int_0^t \mathrm{d}t'\, x_\mathrm{S}(t')\sin\left[\omega_k(t-t')\right] = \sum_k \tilde{B}_k(t) =: \tilde{B}(t), \tag{32}$$

where we have defined

$$\begin{aligned}\delta\Omega_S^2 &:= \sum_k \frac{c_k^2}{\omega_k^2} = \frac{2}{\pi}\int_0^\infty d\omega\frac{\mathcal{J}(\omega)}{\omega} \\ \tilde{B}_k(t) &:= c_k\left(x_k(0)\cos(\omega_k t) + \frac{p_k(0)}{\omega_k}\sin(\omega_k t)\right).\end{aligned} \tag{33}$$

By further defining the dissipation kernel as

$$\chi(t) =: \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega'\,\mathcal{J}(\omega')\sin\left[\omega' t\right]\theta(t). \tag{34}$$

Using this expression—note that it is absolutely important to include the Heaviside in the definition of $\chi(t)$ so that the lower limit of the convolution integral below is $-\infty$ and later on we use the convolution theorem correctly—one can simply rewrite Eq. (32) in the standard Langevin equation form

$$\ddot{x}_\mathrm{S}(t) + (\Omega_\mathrm{S}^2 + \delta\Omega_\mathrm{S}^2)x_\mathrm{S}(t) - \int_{-\infty}^t \mathrm{d}t'\,\chi(t-t')x_\mathrm{S}(t') = \tilde{B}(t). \tag{35}$$

Note that the definition of the bath operators is analogous to Eq. (145), but here the frequencies are the ones from the initial bath, i.e. before the RC mapping. Therefore, the bath correlation function will be analogous to Eq. (79):

$$\sum_k \left\langle \tilde{B}_k(t)\tilde{B}_k(t')\right\rangle = \frac{1}{2}\sum_k \frac{c_k^2}{\omega_k}\left[e^{-i\omega_k(t-t')}\left(\mathcal{N}(\omega_k,T)+1\right) + e^{i\omega_k(t-t')}\mathcal{N}(\omega_k,T)\right] \tag{36}$$

$$= \frac{1}{\pi}\int_0^\infty d\omega\mathcal{J}(\omega)\left[e^{-i\omega(t-t')}\left(\mathcal{N}(\omega,T)+1\right) + e^{i\omega(t-t')}\mathcal{N}(\omega,T)\right]. \tag{37}$$

Finally, by defining the dynamic susceptibility as the Fourier transform of the memory kernel, one can see that

$$\begin{aligned}\chi(\omega) &:= \frac{2}{\pi}\int_0^\infty d\omega' \mathcal{J}(\omega')\int_0^\infty dt e^{i\omega t}\sin\bigl(\omega' t\bigr)\\ &= \frac{2}{\pi}\int_0^\infty d\omega' \mathcal{J}(\omega')\left[\frac{\omega'}{\omega'^2-\omega^2}-\frac{i\pi}{2}[\delta(\omega-\omega')-\delta(\omega+\omega')]\right],\end{aligned} \tag{38}$$

where the real and imaginary parts are connected via the Cramers-Kronig relations. The imaginary part, specifically reads

$$\tilde{\chi}(\omega) := \mathrm{Im}[\chi(\omega)] = \mathcal{J}(\omega)\theta(\omega) - \mathcal{J}(-\omega)\theta(-\omega). \tag{39}$$

On the other hand, we have

$$\begin{aligned}\frac{1}{2}\left\langle\left\{\tilde{B}(\omega),\tilde{B}(\omega')\right\}\right\rangle &= \int\int dt dt' \frac{1}{2}\left\langle\left\{\tilde{B}(t),\tilde{B}(t')\right\}\right\rangle e^{i\omega t}e^{i\omega' t'}\\ &= 2\pi\int_0^\infty d\nu\mathcal{J}(\nu)[(N+1)\delta(\omega-\nu)\delta(\omega'+\nu)+N\delta(\omega+\nu)\delta(\omega'-\nu)\\ &\qquad\qquad +(N+1)\delta(\omega+\nu)\delta(\omega'-\nu)+N\delta(\omega-\nu)\delta(\omega'+\nu)]\\ &= 2\pi\delta(\omega+\omega')\coth(\omega/2T)[\mathcal{J}(\omega)\theta(\omega)-\mathcal{J}(-\omega)\theta(-\omega)]\\ &= 2\pi\delta(\omega+\omega')\coth(\omega/2T)\tilde{\chi}(\omega),\end{aligned} \tag{40}$$

which is the Fluctuation dissipation theorem.

5.1 The steady state

Before finding the general solution of Eq. (35), which requires a Green's function approach, let us solve for its steady state by using the Fourier transformation. Note that, unlike the Laplace transformation, the standard Fourier transform of the convolution requires the bounds in the convolution integral to be $(-\infty,\infty)$. However, in case of Eq. (35) the upper bound is t. That is why we need to let $t\to\infty$ to be able to solve the problem. In this limit, we will have

$$x_{\mathrm{S}}(\omega) = \frac{\tilde{B}(\omega)}{\alpha(\omega)}, \tag{41}$$

where we defined

$$\alpha(\omega) = -\omega^2 + \Omega_S^2 + \delta\Omega_S^2 - \chi(\omega). \tag{42}$$

Note that we have $\alpha(-\omega) = \alpha^*(\omega)$. To see this, first check that the imaginary part of $\chi(\omega)$ is an odd function, i.e., $\tilde{\chi}(-\omega) = -\tilde{\chi}(\omega)$, and thus, according to the Kramers Kronig relation, the real part is even. Then we have

$$\frac{1}{2}\left\langle\{x_{\mathrm{S}}(\omega),x_{\mathrm{S}}(\omega')\}\right\rangle = \frac{2\pi\delta(\omega+\omega')\coth(\omega/2T)\tilde{\chi}(\omega)}{|\alpha(\omega)|^2}. \tag{43}$$

The steady state covariances are given by the inverse Fourier transformation

$$\begin{aligned}
\left\langle x_{\mathrm{S}}^2(t\to\infty)\right\rangle &= \frac{1}{(2\pi)^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\mathrm{d}\omega'\,e^{-it(\omega+\omega')}\frac{1}{2}\left\langle\{x_{\mathrm{S}}(\omega),x_{\mathrm{S}}(\omega')\}\right\rangle \\
&= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\frac{\coth(\omega/2T)\tilde{\chi}(\omega)}{|\alpha(\omega)|^2}, \qquad (44)\\
\left\langle p_{\mathrm{S}}^2(t\to\infty)\right\rangle &= \frac{1}{(2\pi)^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\mathrm{d}\omega'\,e^{-it(\omega+\omega')}\frac{-\omega\omega'}{2}\left\langle\{x_{\mathrm{S}}(\omega),x_{\mathrm{S}}(\omega')\}\right\rangle \\
&= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\frac{\omega^2\coth(\omega/2T)\tilde{\chi}(\omega)}{|\alpha(\omega)|^2}, \qquad (45)\\
\frac{1}{2}\left\langle\{x_{\mathrm{S}}(t\to\infty),p_{\mathrm{S}}(t\to\infty)\}\right\rangle &= \frac{1}{(2\pi)^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\mathrm{d}\omega'\,e^{-it(\omega+\omega')}\frac{i\omega'}{2}\left\langle\{x_{\mathrm{S}}(\omega),x_{\mathrm{S}}(\omega')\}\right\rangle \\
&= \frac{-i}{2\pi}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\frac{\omega\coth(\omega/2T)\tilde{\chi}(\omega)}{|\alpha(\omega)|^2} = 0, \qquad (46)
\end{aligned}$$

where the last equality holds because the integrand is an odd function.

Let us remark that, to calculate both $\langle x^2\rangle$ and $\langle p^2\rangle$ numerically, one can benefit from the fact that the integrands are even functions—which can speed up the integration by only integrating over positive frequencies. Furthermore, note that $\tilde{\chi}(\omega)$ is often decaying beyond the cut-off frequency (either sharply, exponentially, or polynomially depending on the form) and thus one does not need to calculate the integrand far beyond the cut-off. The only important thing, when approximating these integrals with Riemann summation, is that the denominator $\alpha(\omega)$ can have zeros at several points. One has to sample more around these points. However, often, python/matlab built-in integrals take care of this internally.

5.1.1 High temperature limit of the equilibrium covariances

Now, let us study the high temperature limit of Eqs. (44)-(45). Let us first focus on the x-quadrature. Using that $\coth(x)\approx 1/x+\mathcal{O}(x^3)$[Moha]$+\mathcal{O}(x)$ for $x\ll 1$ we get

$$\left\langle x_{\mathrm{S}}^2\right\rangle_{\mathrm{eq}}^{\mathrm{high-T}} \approx \frac{T}{\pi}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\frac{\tilde{\chi}(\omega)}{\omega|\alpha(\omega)|^2}. \qquad (47)$$

[Moha]I see the final result is nice and matches the numerics, so it is correct. But we may need a better justification, because we cannot simply keep only the $1/x$ term; the other terms are also important as the limits of the integral are infinities. Somehow, we are making (potentially justified) assumption that for $\omega\geq T$ we have the integrand vanishing faster than $1/\omega$. Now, we define the Green function as $\mathcal{G}(\omega):=1/\alpha(\omega)$, which matches the definition in Eq. (57), as $\chi(\omega)=\lim_{\eta\to 0^+}\mathcal{L}\{\chi\}(\eta-i\omega)$. Now, define $A(\omega):=-\omega^2+\Omega_{\mathrm{S}}^2+\delta\Omega_{\mathrm{S}}^2-\mathrm{Re}\,\chi(\omega)$ to write

$$\mathcal{G}(\omega)=\frac{1}{A(\omega)-i\tilde{\chi}(\omega)}=\frac{A(\omega)+i\tilde{\chi}(\omega)}{|\alpha(\omega)|^2}, \qquad (48)$$

so that our previous integral reduces to

$$\left\langle x_{\mathrm{S}}^2\right\rangle_{\mathrm{eq}}^{\mathrm{high-T}} \approx \frac{T}{\pi}\int_{-\infty}^{+\infty}\mathrm{d}\omega\,\frac{\mathrm{Im}\{\mathcal{G}(\omega)\}}{\omega}. \qquad (49)$$

Now, using Kramers-Kronig relation

$$\mathrm{Re}\{\mathcal{G}(\omega)\}=\frac{1}{\pi}\mathcal{P}\int_{-\infty}^{+\infty}\mathrm{d}\omega'\,\frac{\mathrm{Im}\{\mathcal{G}(\omega')\}}{\omega'-\omega} \qquad (50)$$

evaluated at $\omega = 0$, we get

$$\langle x_{\mathrm{S}}^2 \rangle_{\mathrm{eq}}^{\mathrm{high-T}} \approx T\,\mathrm{Re}\{\mathcal{G}(0)\} = \frac{T}{\Omega_{\mathrm{S}}^2}, \tag{51}$$

as $\mathrm{Re}\{\chi(0)\} = \delta\Omega_{\mathrm{S}}^2$ [Moha]is this δ supposed to be $1/$? [Erik]no, because it's the real part of the susceptibility, not of the Green function itself. this makes $A(0) = \Omega_{\mathrm{S}}^2$. For the p-quadrature we have

$$\langle p_{\mathrm{S}}^2 \rangle_{\mathrm{eq}}^{\mathrm{high-T}} \approx \frac{T}{\pi} \int_{-\infty}^{+\infty} \mathrm{d}\omega\, \omega\, \mathrm{Im}\{\mathcal{G}(\omega)\}. \tag{52}$$

Now, if we use that

$$G(t) = \frac{2}{\pi} \int_0^{\infty} \mathrm{d}\omega\, \mathrm{Im}\{\mathcal{G}(\omega)\} \sin(\omega t), \quad t > 0 \tag{53}$$

imposing $\dot{G}(t \to 0^+) = 1$ means

$$\frac{2}{\pi} \int_0^{\infty} \mathrm{d}\omega\, \omega\, \mathrm{Im}\{\mathcal{G}(\omega)\} \cos(\omega t) = 1, \tag{54}$$

so that $\langle p_{\mathrm{S}}^2 \rangle_{\mathrm{eq}}^{\mathrm{high-T}} \approx T$.

5.2 Solving the equation with Green functions

Now, let us try to solve Eq. (35) exactly for arbitrary times. To do that, we will refer to the formalism of Green functions. Let us first apply the Laplace transform to Eq. (35) to get

$$\left[s^2 + \left(\Omega_{\mathrm{S}}^2 + \delta\Omega_{\mathrm{S}}^2\right) - \mathcal{L}\{\chi\}(s)\right] \mathcal{L}\{x_{\mathrm{S}}\}(s) = \mathcal{L}\{\tilde{B}\}(s) + s x_{\mathrm{S}}(0) + \dot{x}_{\mathrm{S}}(0), \tag{55}$$

where $\mathcal{L}\{f\}(s)$ denotes the Laplace transform of $f(t)$, that now depends on a new variable s. Here, we have used that $\mathcal{L}\{\ddot{x}_{\mathrm{S}}\} = s^2\mathcal{L}\{x_{\mathrm{S}}\} - s x_{\mathrm{S}}(0) - \dot{x}_{\mathrm{S}}(0)$. Isolating $\mathcal{L}\{x_{\mathrm{S}}\}(s)$ we get

$$\mathcal{L}\{x_{\mathrm{S}}\}(s) = \frac{\mathcal{L}\{\tilde{B}\}(s) + s x_{\mathrm{S}}(0) + \dot{x}_{\mathrm{S}}(0)}{s^2 + \left(\Omega_{\mathrm{S}}^2 + \delta\Omega_{\mathrm{S}}^2\right) - \mathcal{L}\{\chi\}(s)}. \tag{56}$$

Now, if we define the Green function (in conjugate variable space) as

$$\mathcal{G}(s) := \frac{1}{s^2 + \left(\Omega_{\mathrm{S}}^2 + \delta\Omega_{\mathrm{S}}^2\right) - \mathcal{L}\{\chi\}(s)} \tag{57}$$

we can write

$$\mathcal{L}\{x_{\mathrm{S}}\}(s) = \mathcal{G}(s)\mathcal{L}\{\tilde{B}\}(s) + s\mathcal{G}(s) x_{\mathrm{S}}(0) + \mathcal{G}(s)\dot{x}_{\mathrm{S}}(0), \tag{58}$$

which after applying the inverse Laplace transform gives the solution for $x_{\mathrm{S}}(t)$:

$$x_{\mathrm{S}}(t) = x_{\mathrm{S}}(0)\dot{G}(t) + \dot{x}_{\mathrm{S}}(0)G(t) + \int_0^t \mathrm{d}\tau\, G(t-\tau)\tilde{B}(\tau), \tag{59}$$

with the Green functions in the time domain defined via its Laplace transform:

$$\mathcal{L}\{G(t)\} = \frac{1}{s^2 + \left(\Omega_{\mathrm{S}}^2 + \delta\Omega_{\mathrm{S}}^2\right) - \mathcal{L}\{\chi\}(s)}, \tag{60}$$

$$\mathcal{L}\{\dot{G}(t)\} = \frac{s}{s^2 + \left(\Omega_{\mathrm{S}}^2 + \delta\Omega_{\mathrm{S}}^2\right) - \mathcal{L}\{\chi\}(s)}. \tag{61}$$

Now, it is just a matter of computing the Green functions and inserting them into Eq. (59). First, let us compute $\mathcal{L}\{\chi\}(s)$:

$$\mathcal{L}\{\chi\}(s) = \frac{2}{\pi}\int_0^\infty \mathrm{d}\tau\, e^{-s\tau}\int_0^\infty \mathrm{d}\omega\, \mathcal{J}(\omega)\sin(\omega\tau)\Theta(\tau) = \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \mathcal{J}(\omega)\underbrace{\int_0^\infty \mathrm{d}\tau\, e^{-s\tau}\sin(\omega\tau)\Theta(\tau)}_{\frac{\omega}{\omega^2+s^2}} \tag{62}$$

$$= \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \mathcal{J}(\omega)\frac{\omega}{\omega^2+s^2}, \tag{63}$$

where we have defined $\tau := t - t'$. Now, it is just a matter of choosing $\mathcal{J}(\omega)$ and computing the solution. In our case, the spectral density is given by

$$\mathcal{J}(\omega) = \Gamma\frac{\omega}{\omega_m}\sqrt{1-\frac{\omega^2}{\omega_m^2}}\,\Theta(\omega_m-\omega), \tag{64}$$

which yields

$$\mathcal{L}\{\chi\}(s) = \frac{\pi\Gamma\left[s^2\left(2-2\sqrt{\frac{\omega_m^2}{s^2}+1}\right)+\omega_m^2\right]}{4\omega_m^2}. \tag{65}$$

Therefore,

$$\mathcal{L}\{G(t)\} = \frac{1}{s^2+\left(\Omega_\mathrm{S}^2+\delta\Omega_\mathrm{S}^2\right)-\frac{\Gamma\left[s^2\left(2-2\sqrt{\frac{\omega_m^2}{s^2}+1}\right)+\omega_m^2\right]}{2\omega_m^2}}, \tag{66}$$

$$\mathcal{L}\{\dot{G}(t)\} = \frac{s}{s^2+\left(\Omega_\mathrm{S}^2+\delta\Omega_\mathrm{S}^2\right)-\frac{\Gamma\left[s^2\left(2-2\sqrt{\frac{\omega_m^2}{s^2}+1}\right)+\omega_m^2\right]}{2\omega_m^2}}, \tag{67}$$

such that the solution will be given by Eq. (59) with the Green functions defined as the inverse Laplace transforms of Eqs. (66) - (67).

To benchmark the numerical results, let us derive the equilibrium values for the second moments. To do that, we use the fluctuation-dissipation theorem (as in Eq. (40)) to obtain

$$\left\langle x^2\right\rangle_\mathrm{eq} = -\frac{1}{\pi}\int_0^{\omega_m}\mathrm{d}\omega\,\tilde{\chi}(\omega)\coth\left(\frac{\beta\omega}{2}\right) \tag{68}$$

$$\left\langle p^2\right\rangle_\mathrm{eq} = -\frac{1}{\pi}\int_0^{\omega_m}\mathrm{d}\omega\,\omega^2\tilde{\chi}(\omega)\coth\left(\frac{\beta\omega}{2}\right) \tag{69}$$

6 Approach 2: (local) GKLS master equation without reaction coordinate

Now, let us derive the GKLS master equation for the system interacting with an harmonic bath. To do that, recall the Hamiltonian in Eq. (22), so we can write (in the interaction picture)

$$\tilde{H}_{\mathrm{SB}} = e^{i(H_{\mathrm{S}}+H_{\mathrm{B}})t} H_{\mathrm{SB}} e^{-i(H_{\mathrm{S}}+H_{\mathrm{B}})t}, \tag{70}$$

so that the dynamics will be given by

$$\frac{\partial \tilde{\rho}(t)}{\partial t} = -i\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{SB}}(t), \tilde{\rho}(0)\right] - \int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{SB}}(t), \left[\tilde{H}_{\mathrm{SB}}(s), \tilde{\rho}(s)\right]\right]. \tag{71}$$

Rewriting the interaction Hamiltonian (in the Schrödinger picture) as $H_{\mathrm{SB}} = A \otimes B$, with $A := x_{\mathrm{S}}$ and $B := -\sum_k c_k x_k$, we note that the first term vanishes, as $\mathrm{Tr}_B(\rho_{\mathrm{B}}\tilde{B}(t)) = 0$. For the second term we have:

$$\frac{\partial \tilde{\rho}(t)}{\partial t} = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{SB}}(t), \left[\tilde{H}_{\mathrm{SB}}(s), \tilde{\rho}(s)\right]\right] = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{SB}}(t), \left[\tilde{H}_{\mathrm{SB}}(t-s), \tilde{\rho}_{\mathrm{S}}(t-s)\otimes\tilde{\rho}_B\right]\right] \tag{72}$$

$$= -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{SB}}(t)\tilde{H}_{\mathrm{SB}}(t-s)\tilde{\rho}_{\mathrm{S}}(t-s)\otimes\tilde{\rho}_B - \tilde{H}_{\mathrm{SB}}(t-s)\tilde{\rho}_{\mathrm{S}}(t-s)\otimes\tilde{\rho}_B\tilde{H}_{\mathrm{SB}}(t) + \mathrm{h.c.}\right], \tag{73}$$

where we swapped $s \leftrightarrow t-s$. Now, we need to transform the operators A and B into the interaction picture:

$$A = \frac{1}{\sqrt{2\Omega_{\mathrm{S}}}}(a + a^\dagger) \Rightarrow \tilde{A}(t) = \frac{1}{\sqrt{2\Omega_{\mathrm{S}}}}(e^{-i\Omega_{\mathrm{S}}t}a + e^{i\Omega_{\mathrm{S}}t}a^\dagger), \tag{74}$$

$$B = -\frac{1}{\sqrt{2\omega_k}}\sum_k c_k(b_k + b_k^\dagger) \Rightarrow \tilde{B} = -\frac{1}{\sqrt{2\omega_k}}\sum_k c_k(e^{-i\omega_k t}b_k + e^{i\omega_k t}b_k^\dagger). \tag{75}$$

Substituting the explicit form of the Hamiltonian we get:

$$\frac{\partial \tilde{\rho}(t)}{\partial t} = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\Big[(\tilde{A}(t)\otimes\tilde{B}(t))(\tilde{A}(t-s)\otimes\tilde{B}(t-s))\tilde{\rho}_{\mathrm{S}}(t-s)\otimes\tilde{\rho}_B \tag{76}$$

$$- (\tilde{A}(t-s)\otimes\tilde{B}(t-s))\tilde{\rho}_{\mathrm{S}}(t-s)\otimes\tilde{\rho}_B(\tilde{A}(t)\otimes\tilde{B}(t)) + \mathrm{h.c.}\Big] \tag{77}$$

$$= -\int_0^t \mathrm{d}s\left(\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_{\mathrm{S}}(t-s) - \tilde{A}(t-s)\tilde{\rho}_{\mathrm{S}}(t-s)\tilde{A}(t)\right)\mathrm{Tr}_B\left(\tilde{B}(t)\tilde{B}(t-s)\tilde{\rho}_B\right) + \mathrm{h.c.} \tag{78}$$

To simplify the expression, let us compute the bath-dependent term, which can be shown to be

$$\mathrm{Tr}_B\left(\tilde{B}(t)\tilde{B}(t-s)\tilde{\rho}_B\right) = \frac{1}{2}\sum_k \frac{c_k^2}{\omega_k}\left[e^{-i\omega_k s}\left(\mathcal{N}(\omega_k,T)+1\right) + e^{i\omega_k s}\mathcal{N}(\omega_k,T)\right], \tag{79}$$

where $\mathcal{N}(\omega_k, T)$ is the bosonic occupation number of the mode with associated frequency ω_k. Let us then simplify the expression:

$$
\begin{aligned}
\frac{\partial \tilde{\rho}(t)}{\partial t} &= -\frac{1}{2}\sum_k \frac{c_k^2}{\omega_k}\int_0^t \mathrm{d}s \left(\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_{\mathrm{S}}(t) - \tilde{A}(t-s)\tilde{\rho}_{\mathrm{S}}(t)\tilde{A}(t)\right) \times && (80)\\
&\times \left[e^{-i\omega_k s}\left(\mathcal{N}(\omega_k, T)+1\right) + e^{i\omega_k s}\mathcal{N}(\omega_k, T)\right] && (81)\\
&= \frac{1}{4\Omega_{\mathrm{S}}}\sum_k \frac{c_k^2}{\omega_k}\int_0^\infty \mathrm{d}s \Big[e^{-i\omega_{\mathrm{S}}(2t-s)}a\tilde{\rho}_{\mathrm{S}}(t)a + e^{i\Omega_{\mathrm{S}}s}a\tilde{\rho}_{\mathrm{S}}(t)a^\dagger + e^{-i\Omega_{\mathrm{S}}s}a^\dagger\tilde{\rho}_{\mathrm{S}}(t)a + && (82)\\
&+ e^{i\Omega_{\mathrm{S}}(2t-s)}a^\dagger\tilde{\rho}_{\mathrm{S}}(t)a^\dagger - \Big(e^{-i\Omega_{\mathrm{S}}(2t-s)}a^2\tilde{\rho}_{\mathrm{S}}(t) + e^{i\Omega_{\mathrm{S}}s}a^\dagger a\tilde{\rho}_{\mathrm{S}}(t) + e^{-i\Omega_{\mathrm{S}}s}aa^\dagger\tilde{\rho}_{\mathrm{S}}(t) + && (83)\\
&+ e^{i\Omega_{\mathrm{S}}(2t-s)}(a^\dagger)^2\tilde{\rho}_{\mathrm{S}}(t)\Big)\Big] \left[e^{-i\omega_k s}\left(\mathcal{N}(\omega_k, T)+1\right) + e^{i\omega_k s}\mathcal{N}(\omega_k, T)\right]. && (84)
\end{aligned}
$$

If we expand the product and group the phases we get:

$$
\begin{aligned}
\frac{\partial \tilde{\rho}(t)}{\partial t} &= \frac{1}{4\Omega_{\mathrm{S}}}\sum_k \frac{c_k^2}{\omega_k}\int_0^\infty \mathrm{d}s && (85)\\
&\Big[e^{-2i\Omega_{\mathrm{S}}t}\left(a\tilde{\rho}_{\mathrm{S}}(t)a - a^2\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega_k-\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega_k, T)+1) + e^{i(\omega_k+\Omega_{\mathrm{S}})s}\mathcal{N}(\omega_k, T)\right) + && (86)\\
&+ \left(a\tilde{\rho}_{\mathrm{S}}(t)a^\dagger - a^\dagger a\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega_k-\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega_k, T)+1) + e^{i(\omega_k+\Omega_{\mathrm{S}})s}\mathcal{N}(\omega_k, T)\right) + && (87)\\
&+ \left(a^\dagger\tilde{\rho}_{\mathrm{S}}(t)a - aa^\dagger\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega_k+\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega_k, T)+1) + e^{i(\omega_k-\Omega_{\mathrm{S}})s}\mathcal{N}(\omega_k, T)\right) + && (88)\\
&+ e^{2i\Omega_{\mathrm{S}}t}\left(a^\dagger\tilde{\rho}_{\mathrm{S}}(t)a^\dagger - (a^\dagger)^2\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega_k+\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega_k, T)+1) + e^{i(\omega_k-\Omega_{\mathrm{S}})s}\mathcal{N}(\omega_k, T)\right)\Big] && (89)
\end{aligned}
$$

Now, because of the secular approximation, we can neglect the terms containing $e^{\pm 2i\Omega_{\mathrm{S}}t}$, as those are the terms containing a subtraction of different system frequencies [**?**], which in our case are $\pm\Omega_{\mathrm{S}}$, as they are the frequencies associated to the rotation of the operators into the interaction picture in Eq. (74). Then, the master equation reads

$$
\begin{aligned}
\frac{\partial \tilde{\rho}(t)}{\partial t} &= \frac{1}{4\Omega_{\mathrm{S}}}\sum_k \frac{c_k^2}{\omega_k}\int_0^\infty \mathrm{d}s && (90)\\
&\Big[\left(a\tilde{\rho}_{\mathrm{S}}(t)a^\dagger - a^\dagger a\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega_k-\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega_k, T)+1) + e^{i(\omega_k+\Omega_{\mathrm{S}})s}\mathcal{N}(\omega_k, T)\right) + && (91)\\
&+ \left(a^\dagger\tilde{\rho}_{\mathrm{S}}(t)a - aa^\dagger\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega_k+\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega_k, T)+1) + e^{i(\omega_k-\Omega_{\mathrm{S}})s}\mathcal{N}(\omega_k, T)\right)\Big] && (92)
\end{aligned}
$$

Using that $\mathcal{J}(\omega) = \frac{\pi}{2}\sum_k \frac{c_k^2}{\omega_k}\delta(\omega-\omega_k)$ we can rewrite our expression as follows:

$$
\begin{aligned}
\frac{\partial \tilde{\rho}(t)}{\partial t} &= \frac{1}{2\pi\Omega_{\mathrm{S}}}\int_0^\infty \mathrm{d}\omega\, \mathcal{J}(\omega)\int_0^\infty \mathrm{d}s && (93)\\
&\Big[\left(a\tilde{\rho}_{\mathrm{S}}(t)a^\dagger - a^\dagger a\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega-\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega, T)+1) + e^{i(\omega+\Omega_{\mathrm{S}})s}\mathcal{N}(\omega, T)\right) + && (94)\\
&+ \left(a^\dagger\tilde{\rho}_{\mathrm{S}}(t)a - aa^\dagger\tilde{\rho}_{\mathrm{S}}(t)\right)\left(e^{-i(\omega+\Omega_{\mathrm{S}})s}(\mathcal{N}(\omega, T)+1) + e^{i(\omega-\Omega_{\mathrm{S}})s}\mathcal{N}(\omega, T)\right)\Big] && (95)
\end{aligned}
$$

Now, using that

$$
\begin{aligned}
\int_0^\infty \mathrm{d}s\, e^{\pm i(\omega-\Omega_{\mathrm{S}})s} &= \pi\delta(\omega-\Omega_{\mathrm{S}}) \pm i\mathbb{P}\frac{1}{\omega-\Omega_{\mathrm{S}}} && (96)\\
\int_0^\infty \mathrm{d}s\, e^{\pm i(\omega+\Omega_{\mathrm{S}})s} &= \pi\delta(\omega+\Omega_{\mathrm{S}}) \pm i\mathbb{P}\frac{1}{\omega+\Omega_{\mathrm{S}}} && (97)
\end{aligned}
$$

we can write:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{2\Omega_{\rm S}}\left[a\tilde{\rho}(t)a^\dagger\left(\mathcal{J}(\Omega_{\rm S})(\mathcal{N}(\Omega_{\rm S},T)+1) - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)(\mathcal{N}+1)}{\omega-\Omega_{\rm S}} + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)\mathcal{N}}{\omega+\Omega_{\rm S}}\right)+\right. \tag{98}$$

$$+ a^\dagger\tilde{\rho}(t)a\left(\mathcal{J}(\Omega_{\rm S})\mathcal{N}(\Omega_{\rm S},T) + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)\mathcal{N}}{\omega-\Omega_{\rm S}} - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)(\mathcal{N}+1)}{\omega+\Omega_{\rm S}}\right)+ \tag{99}$$

$$- a^\dagger a\tilde{\rho}(t)\left(\mathcal{J}(\Omega_{\rm S})(\mathcal{N}(\Omega_{\rm S},T)+1) - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)(\mathcal{N}+1)}{\omega-\Omega_{\rm S}} + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)\mathcal{N}}{\omega+\Omega_{\rm S}}\right)+ \tag{100}$$

$$\left.- aa^\dagger\tilde{\rho}(t)\left(\mathcal{J}(\Omega_{\rm S})\mathcal{N}(\Omega_{\rm S},T) + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)\mathcal{N}}{\omega-\Omega_{\rm S}} - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}(\omega)(\mathcal{N}+1)}{\omega+\Omega_{\rm S}}\right)\right], \tag{101}$$

where we write $\tilde{\rho}(t) := \tilde{\rho}_{\rm S}(t)$ and $\mathcal{N} := \mathcal{N}(\omega,T)$ to shorten the notation (we reincorporate all the labels later). When reincorporating the hermitian conjugate, we can write the previous expression in a more familiar way:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{\Omega_{\rm S}}\mathcal{J}(\Omega_{\rm S})\left[\mathcal{N}(\Omega_{\rm S},T)\left(a^\dagger\tilde{\rho}a - \frac{1}{2}\left\{aa^\dagger,\tilde{\rho}\right\}\right) + (\mathcal{N}(\Omega_{\rm S},T)+1)\left(a\tilde{\rho}a^\dagger - \frac{1}{2}\left\{a^\dagger a,\tilde{\rho}\right\}\right)\right] \tag{102}$$

$$+ \text{Lamb shift terms}, \tag{103}$$

from where we can identify the following Lindblad operators:

$$L_1 = \sqrt{\frac{\mathcal{J}(\Omega_{\rm S})\left(\mathcal{N}(\Omega_S,T)+1\right)}{\Omega_S}}a, \tag{104}$$

$$L_2 = \sqrt{\frac{\mathcal{J}(\Omega_{\rm S})\mathcal{N}(\Omega_S,T)}{\Omega_S}}a^\dagger. \tag{105}$$

With these operators, we can compute the evolution of the first and second moments as in [**?**], which is explicitly given by the following set of equations:

$$\frac{\mathrm{d}\langle R\rangle}{\mathrm{d}t} = A\langle R\rangle, \tag{106}$$

$$\frac{\mathrm{d}\sigma}{\mathrm{d}t} = A\sigma + \sigma A^{\rm T} + D, \tag{107}$$

with $R := \left(x_{\rm S}, p_{\rm S}\right)^T$ being the vector of quadratures. Moreover, the matrices A and D are defined as follows:

$$A := -i\Omega\left[G - \mathrm{Im}(CC^\dagger)\right] \tag{108}$$

$$D := \Omega\,\mathrm{Re}(CC^\dagger)\Omega \tag{109}$$

with

$$\Omega := i\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}, \quad G := \begin{pmatrix}\Omega_{\rm S}^2 & 0\\ 0 & 1\end{pmatrix}, \quad C := (c_1^{\rm T}; c_2^{\rm T})^{\rm T}, \tag{110}$$

where c_k is defined such that the Lindblad operators are given by $L_k = c_k^{\rm T}R$. Therefore, the vectors c_k, with $k\in\{1,2\}$, are given by

$$c_1 = \sqrt{\frac{\mathcal{J}(\Omega_{\rm S})\mathcal{N}(\Omega_{\rm S},T)}{2\Omega_{\rm S}}}\begin{pmatrix}\sqrt{\Omega_{\rm S}}\\ -i/\sqrt{\Omega_{\rm S}}\end{pmatrix}, \qquad c_2 = \sqrt{\frac{\mathcal{J}(\Omega_{\rm S})\left(\mathcal{N}(\Omega_{\rm S},T)+1\right)}{2\Omega_{\rm S}}}\begin{pmatrix}\sqrt{\Omega_{\rm S}}\\ i/\sqrt{\Omega_{\rm S}}\end{pmatrix}. \tag{111}$$

With all these definitions, we can now compute A and D. Let us start by explicitly writing down C:

$$C = \begin{pmatrix} \sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{S}})\mathcal{N}(\Omega_{\mathrm{S}},T)}{2}} & \sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{S}})(\mathcal{N}(\Omega_{\mathrm{S}},T)+1)}{2}} \\ -\frac{i}{\Omega_S}\sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{S}})\mathcal{N}(\Omega_{\mathrm{S}},T)}{2}} & \frac{i}{\Omega_{\mathrm{S}}}\sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{S}})(\mathcal{N}(\Omega_{\mathrm{S}},T)+1)}{2}}. \end{pmatrix} \tag{112}$$

Now, it can be shown that the matrices A and D are given by

$$A = \begin{pmatrix} -\frac{\mathcal{J}(\Omega_{\mathrm{S}})}{2\Omega_{\mathrm{S}}} & 1 \\ -\Omega_{\mathrm{S}}^2 & -\frac{\mathcal{J}(\Omega_{\mathrm{S}})}{2\Omega_{\mathrm{S}}} \end{pmatrix}, \quad D = \begin{pmatrix} \frac{\mathcal{J}(\Omega_{\mathrm{S}})}{\Omega_{\mathrm{S}}^2}\left(\mathcal{N}(\Omega_{\mathrm{S}},T)+\frac{1}{2}\right) & 0 \\ 0 & \mathcal{J}(\Omega_{\mathrm{S}})\left(\mathcal{N}(\Omega_{\mathrm{S}},T)+\frac{1}{2}\right) \end{pmatrix}. \tag{113}$$

This yields the following equilibrium covariance matrix, σ_{eq}:

$$\sigma_{\mathrm{eq}} = \begin{pmatrix} \frac{1}{\Omega_{\mathrm{S}}}\left(\mathcal{N}(\Omega_{\mathrm{S}},T)+\frac{1}{2}\right) & 0 \\ 0 & \Omega_{\mathrm{S}}\left(\mathcal{N}(\Omega_{\mathrm{S}},T)+\frac{1}{2}\right), \end{pmatrix} \tag{114}$$

which is consistent with the theoretical predictions.

7 Approach 3: (global) master equation for the system-reaction coordinate unit

7.1 Reaction coordinate (RC) mapping

In this last approach, we want to extract a collective coordinate of the bath, named reaction coordinate (RC), such that it is the one coupling to the system. Therefore,

$$X_{\mathrm{RC}} := \frac{1}{\lambda}\sum_k c_k x_k. \tag{115}$$

This can be implemented by a transformation of the form

$$X_i = \sum_j O_{ij} x_j, \tag{116}$$

$$P_i = \sum_j O_{ij} p_j. \tag{117}$$

Note that the transformation needs to be the same for both operators to preserve the canonical commutation relation:

$$[X_i, P_j] = \sum_{m,l}[O_{im}x_m, O_{jl}p_l] = \sum_{m,l} O_{im}O_{jl}\underbrace{[x_m, p_l]}_{i\delta_{ml}} = i\sum_{m,l} O_{im}O_{jl}\delta_{ml} = i\sum_m O_{im}O_{jm} = i\delta_{ij}, \tag{118}$$

as O is an orthogonal matrix ($O^{\mathrm{T}} = O^{-1}$). If the transformation is such that the residual baths also have a normal form, the following relation holds:

$$\sum_k O_{ik}\omega_k^2 O_{jk} = \delta_{ij}\Omega_i^2, \tag{119}$$

with $i, j > 1$, since the first row of the transformation is fixed by Eq. (115). Using these insights, we can rewrite the Hamiltonian as a function of the RC operators:

$$H = \frac{1}{2}(p_S^2+\Omega_S^2 x_S^2)+\frac{1}{2}\sum_k \frac{c_k^2}{\omega_k^2}x_S^2 - x_S\lambda X_{\mathrm{RC}} + \frac{1}{2}(P_{\mathrm{RC}}^2+\Omega_{\mathrm{RC}}^2 X_{\mathrm{RC}}^2) - X_{\mathrm{RC}}\sum_{k>1} C_k X_k + \frac{1}{2}\sum_{k>1}(P_k^2+\Omega_k^2 X_k^2), \tag{120}$$

with $C_k = -\sum_m \omega_m^2 O_{m1} O_{mk}$. Observing that $O_{1k} = c_k/\lambda$ we can derive a constraint for λ from the new canonical commutation relations for the RC operators:

$$[X_{\mathrm{RC}}, P_{\mathrm{RC}}] = \sum_{m,l}[O_{1m}x_m, O_{1l}p_l] = \sum_{m,l} O_{1m}O_{1l}[x_m, p_l] = i\sum_{m,l} O_{1m}O_{1l}\delta_{ml} = i\sum_m O_{1m}^2 \tag{121}$$

$$= \frac{i}{\lambda^2}\sum_m c_m^2 = i \Rightarrow \lambda^2 = \sum_k c_k^2. \tag{122}$$

Now, let us relate all these quantities to the original spectral density, such that we can know the transformation by just knowing the original spectral density. To do that, note from Eq. (119) that the RC frequency fulfills

$$\Omega_{\mathrm{RC}}^2 = \sum_k \omega_k^2 O_{1k}^2 = \frac{1}{\lambda^2}\sum_k c_k^2\omega_k^2. \tag{123}$$

by using the definition if the spectral density (31) the following relations hold:

$$\delta\Omega_{\mathrm{S}}^2 := \sum_k \frac{c_k^2}{\omega_k^2} = \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{\mathcal{J}(\omega)}{\omega}, \tag{124}$$

$$\lambda^2 = \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \omega \mathcal{J}(\omega), \tag{125}$$

$$\Omega_{\mathrm{RC}}^2 = \frac{2}{\pi\lambda^2}\int_0^\infty \mathrm{d}\omega\, \omega^3 \mathcal{J}(\omega). \tag{126}$$

Now, it is left to relate $\mathcal{J}(\omega)$ with the spectral density of the RC, $\mathcal{J}_{\mathrm{RC}}(\omega)$. To do that, refer to Section 7.5 to obtain the following expression [**?**, **?**, **?**, **?**]:

$$\mathcal{J}_{\mathrm{RC}}(\omega) = \frac{\lambda^2 \mathcal{J}(\omega)}{|W^+(\omega)|^2}, \tag{127}$$

where $W^+(\omega)$ is the Cauchy transform of the original spectral density, which is given by

$$W(z) = \frac{1}{\pi}\int_{-\infty}^{+\infty} \mathrm{d}\omega\, \frac{\mathcal{J}(\omega)}{\omega - z} \tag{128}$$

with

$$W^+(\omega) := \lim_{\epsilon \searrow 0} W(\omega + i\epsilon), \quad \omega \in \mathbb{R}. \tag{129}$$

Therefore, the new RC spectral density takes the form

$$\mathcal{J}_{\mathrm{RC}}(\omega) = \frac{\pi}{2}\sum_k \frac{C_k^2}{\Omega_k}\delta(\omega - \Omega_k), \tag{130}$$

and the following relation also holds:

$$\frac{\lambda^2}{\delta\Omega_{\mathrm{S}}^2} = \Omega_{\mathrm{RC}}^2 - \delta\Omega_{\mathrm{RC}}^2, \tag{131}$$

with

$$\delta\Omega_{\mathrm{RC}}^2 := \sum_k \frac{C_k^2}{\Omega_k^2}. \tag{132}$$

Note that this will determine the physical frequency of the reaction coordinate, as $\lambda^2/\delta\Omega_{\mathrm{S}}^2$ is the factor multiplying the RC position operator. For this reason, let us define the *physical* reaction coordinate frequency, $\Omega_{\mathrm{RC}}^{\mathrm{phy}}$, as

$$\Omega_{\mathrm{RC}}^{\mathrm{phy}} := \frac{\lambda}{\delta\Omega_{\mathrm{S}}} = \sqrt{\Omega_{\mathrm{RC}}^2 - \delta\Omega_{\mathrm{RC}}^2}. \tag{133}$$

Using Eq. (130) we can rewrite the total Hamiltonian as

$$H = \frac{1}{2}(p_S^2 + \Omega_S^2 x_S^2) + \frac{1}{2}\left[P_{\mathrm{RC}}^2 + \frac{\lambda^2}{\delta\Omega_{\mathrm{S}}^2}\left(X_{\mathrm{RC}} - \frac{\delta\Omega_{\mathrm{S}}^2}{\lambda}s\right)^2\right] + \frac{1}{2}\sum_k\left[P_k^2 + \Omega_k^2\left(X_k - \frac{C_k}{\Omega_k^2}X_{\mathrm{RC}}\right)^2\right] \tag{134}$$

which is in Brownian form and reproduces (up to the shift in the system Hamiltonian) the RC Hamiltonian in [**?**].

7.2 Master equation for the system-RC

Now, we would like to derive a global master equation for the system-reaction coordinate unit. To this aim, we should find the normal modes of the composite Hamiltonian. Note that the original Hamiltonian has frequencies $\{\Omega_1^2 = \Omega_S^2 + \delta\Omega_S^2, \Omega_2^2 = \Omega_{\rm RC}^2\}$ and has a coupling $-\lambda x_S x_{\rm RC}$. Luckily, the interaction term is simple enough to obtain the uncoupled quadratures $\{u_+, u_-\}$ with the corresponding normal mode frequencies

$$\Omega_\pm^2 = \frac{1}{2}\left(\Omega_1^2 + \Omega_2^2 \pm \sqrt{4\lambda^2 + (\Omega_1^2 - \Omega_2^2)^2}\right), \tag{135}$$

and free quadratures

$$\begin{pmatrix} u_+ \\ u_- \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} x_S \\ x_{\rm RC} \end{pmatrix}, \tag{136}$$

with

$$\cos^2(\theta) = \frac{\Omega_1^2 - \Omega_2^2 + \sqrt{4\lambda^2 + (\Omega_1^2 - \Omega_2^2)^2}}{2\sqrt{4\lambda^2 + (\Omega_1^2 - \Omega_2^2)^2}}. \tag{137}$$

The corresponding 4 jump operators, $\{b_+, b_-, b_+^\dagger, b_-^\dagger\}$ are given by

$$\begin{aligned}
\begin{pmatrix} b_+ \\ b_+^\dagger \\ b_- \\ b_-^\dagger \end{pmatrix} &= \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{\Omega_+} & \frac{i}{\sqrt{\Omega_+}} & 0 & 0 \\ \sqrt{\Omega_+} & \frac{-i}{\sqrt{\Omega_+}} & 0 & 0 \\ 0 & 0 & \sqrt{\Omega_-} & \frac{i}{\sqrt{\Omega_-}} \\ 0 & 0 & \sqrt{\Omega_-} & \frac{-i}{\sqrt{\Omega_-}} \end{pmatrix} \begin{pmatrix} u_+ \\ \dot{u}_+ \\ u_- \\ \dot{u}_- \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{\Omega_+} & \frac{i}{\sqrt{\Omega_+}} & 0 & 0 \\ \sqrt{\Omega_+} & \frac{-i}{\sqrt{\Omega_+}} & 0 & 0 \\ 0 & 0 & \sqrt{\Omega_-} & \frac{i}{\sqrt{\Omega_-}} \\ 0 & 0 & \sqrt{\Omega_-} & \frac{-i}{\sqrt{\Omega_-}} \end{pmatrix} \begin{pmatrix} \cos(\theta) & 0 & -\sin(\theta) & 0 \\ 0 & \cos(\theta) & 0 & -\sin(\theta) \\ \sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & \sin(\theta) & 0 & \cos(\theta) \end{pmatrix} \begin{pmatrix} x_S \\ p_S \\ x_{\rm RC} \\ p_{\rm RC} \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{\Omega_+}\cos(\theta) & \frac{i\cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+}\sin(\theta) & \frac{-i\sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_+}\cos(\theta) & \frac{-i\cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+}\sin(\theta) & \frac{i\sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_-}\sin(\theta) & \frac{i\sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-}\cos(\theta) & \frac{i\cos(\theta)}{\sqrt{\Omega_-}} \\ \sqrt{\Omega_-}\sin(\theta) & \frac{-i\sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-}\cos(\theta) & \frac{-i\cos(\theta)}{\sqrt{\Omega_-}} \end{pmatrix} \begin{pmatrix} x_S \\ p_S \\ x_{\rm RC} \\ p_{\rm RC} \end{pmatrix}.
\end{aligned} \tag{138}$$

Now, let us try to derive the (global) master equation (ME) for the system + RC. To do that, let us first identify some terms in the Hamiltonian as

$$\begin{aligned}
H = &\underbrace{\frac{1}{2}(p_S^2 + \Omega_S^2 x_S^2) + \frac{1}{2}\left[P_{\rm RC}^2 + \frac{\lambda^2}{\delta\Omega_{\rm S}^2}\left(X_{\rm RC} - \frac{\delta\Omega_{\rm S}^2}{\lambda}x_S\right)^2\right]}_{H_{\rm S+RC}} + \underbrace{\frac{1}{2}\sum_k \left(P_k^2 + \Omega_k^2 X_k^2\right)}_{H_{\rm RB}} \\
&\underbrace{-\sum_k X_k C_k X_{\rm RC}}_{H_{\rm I}} + \underbrace{\frac{1}{2}\sum_k \frac{C_k^2}{\Omega_k^2} X_{\rm RC}^2}_{H_{\rm S+RC}},
\end{aligned} \tag{139}$$

so we can write the interacting Hamiltonian in the interaction picture as

$$\tilde{H}_{\rm I}(t) = e^{i(H_{\rm S+RC} + H_{\rm RB})t} H_{\rm I} e^{-i(H_{\rm S+RC} + H_{\rm RB})t}. \tag{140}$$

As before, the dynamics of the system in the interaction picture reads:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_\mathrm{I}(t),\left[\tilde{H}_\mathrm{I}(s),\tilde{\rho}(s)\right]\right] = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_\mathrm{I}(t),\left[\tilde{H}_\mathrm{I}(t-s),\tilde{\rho}_\mathrm{S+RC}(t-s)\otimes\tilde{\rho}_B\right]\right] \tag{141}$$

$$= -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_\mathrm{I}(t)\tilde{H}_\mathrm{I}(t-s)\tilde{\rho}_\mathrm{S+RC}(t-s)\otimes\tilde{\rho}_B - \tilde{H}_\mathrm{I}(t-s)\tilde{\rho}_\mathrm{S+RC}(t-s)\otimes\tilde{\rho}_B\tilde{H}_\mathrm{I}(t) + \text{h.c.}\right] \tag{142}$$

$$= -\int_0^t \mathrm{d}s\left(\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_\mathrm{S+RC}(t-s) - \tilde{A}(t-s)\tilde{\rho}_\mathrm{S+RC}(t-s)\tilde{A}(t)\right)\mathrm{Tr}_B\left(\tilde{B}(t)\tilde{B}(t-s)\tilde{\rho}_B\right) + \text{h.c.}, \tag{143}$$

where we again changed $s \leftrightarrow t-s$ and defined:

$$A := X_\mathrm{RC} = \frac{b_- + b_-^\dagger}{\sqrt{2\Omega_-}}\cos\theta - \frac{b_+ + b_+^\dagger}{\sqrt{2\Omega_+}}\sin\theta \Rightarrow \tilde{A}(t) = \frac{b_- e^{-i\Omega_- t} + b_-^\dagger e^{i\Omega_- t}}{\sqrt{2\Omega_-}}\cos\theta - \frac{b_+ e^{-i\Omega_+ t} + b_+^\dagger e^{i\Omega_+ t}}{\sqrt{2\Omega_+}}\sin\theta, \tag{144}$$

$$B = -\frac{1}{\sqrt{2\Omega_k}}\sum_k C_k(b_k + b_k^\dagger) \Rightarrow \tilde{B}(t) = -\frac{1}{\sqrt{2\Omega_k}}\sum_k C_k(e^{-i\Omega_k t}b_k + e^{i\Omega_k t}b_k^\dagger). \tag{145}$$

so that

$$\mathrm{Tr}_B\left(\tilde{B}(t)\tilde{B}(t-s)\tilde{\rho}_B\right) = \frac{1}{2}\sum_k \frac{C_k^2}{\Omega_k}\left[e^{-i\Omega_k s}\left(\mathcal{N}(\Omega_k,T)+1\right) + e^{i\Omega_k s}\mathcal{N}(\Omega_k,T)\right]. \tag{146}$$

Let us first look at the term $\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_\mathrm{S+RC}(t-s)$:

$$\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_\mathrm{S+RC}(t-s) = \left(\frac{b_- e^{-i\Omega_- t} + b_-^\dagger e^{i\Omega_- t}}{\sqrt{2\Omega_-}}\cos\theta - \frac{b_+ e^{-i\Omega_+ t} + b_+^\dagger e^{i\Omega_+ t}}{\sqrt{2\Omega_+}}\sin\theta\right) \tag{147}$$

$$\times\left(\frac{b_- e^{-i\Omega_-(t-s)} + b_-^\dagger e^{i\Omega_-(t-s)}}{\sqrt{2\Omega_-}}\cos\theta - \frac{b_+ e^{-i\Omega_+(t-s)} + b_+^\dagger e^{i\Omega_+(t-s)}}{\sqrt{2\Omega_+}}\sin\theta\right)\tilde{\rho}_\mathrm{S+RC}(t-s) \tag{148}$$

$$= \left[(1)+(2)+(3)+(4)\right]\tilde{\rho}_\mathrm{S+RC}(t-s), \tag{149}$$

with

$$(1) = \frac{\sin^2\theta}{2\Omega_+}\left(b_+b_+^\dagger e^{-i\Omega_+ s} + b_+^\dagger b_+ e^{i\Omega_+ s} + b_+^2 e^{-i\Omega_+(2t-s)} + b_+^{\dagger 2} e^{i\Omega_+(2t-s)}\right) \tag{150}$$

$$(2) = -\frac{\sin\theta\cos\theta}{2\sqrt{\Omega_+\Omega_-}}\left(b_+b_- e^{i\Omega_- s}e^{-i(\Omega_+ + \Omega_-)t} + b_+^\dagger b_- e^{i\Omega_- s}e^{i(\Omega_+ - \Omega_-)t} + b_+ b_-^\dagger e^{-i\Omega_- s}e^{-i(\Omega_+ - \Omega_-)t}\right. \tag{151}$$

$$\left. + b_+^\dagger b_-^\dagger e^{-i\Omega_- s}e^{i(\Omega_+ + \Omega_-)t}\right) \tag{152}$$

$$(3) = -\frac{\sin\theta\cos\theta}{2\sqrt{\Omega_+\Omega_-}}\left(b_-b_+ e^{i\Omega_+ s}e^{-i(\Omega_+ + \Omega_-)t} + b_-^\dagger b_+ e^{i\Omega_+ s}e^{-i(\Omega_+ - \Omega_-)t} + b_- b_+^\dagger e^{-i\Omega_+ s}e^{i(\Omega_+ - \Omega_-)t}\right. \tag{153}$$

$$\left. + b_-^\dagger b_+^\dagger e^{-i\Omega_+ s}e^{i(\Omega_+ + \Omega_-)t}\right) \tag{154}$$

$$(4) = \frac{\cos^2\theta}{2\Omega_-}\left(b_-b_-^\dagger e^{-i\Omega_- s} + b_-^\dagger b_- e^{i\Omega_- s} + b_-^2 e^{-i\Omega_-(2t-s)} + b_-^{\dagger 2} e^{i\Omega_-(2t-s)}\right) \tag{155}$$

Terms (1) and (4) will give the usual time-independent dissipation terms in the GKLS master equation. In our case, we want to go a step further and also consider the terms with subtraction of frequencies, i.e. exponentials with $\Omega_+ - \Omega_-$. In that regime, (2) + (3) becomes

$$-\frac{\sin\theta\cos\theta}{2\sqrt{\Omega_+\Omega_-}}\Big\{e^{i(\Omega_+-\Omega_-)t}\Big[\Big(b_+^\dagger b_-\tilde{\rho}-b_-\tilde{\rho}b_+^\dagger\Big)\Big(e^{-i(\Omega_k-\Omega_-)s}(\mathcal{N}(\Omega_k,T)+1)+e^{i(\Omega_k+\Omega_-)s}\mathcal{N}(\Omega_k,T)\Big) \tag{156}$$

$$+\ \Big(b_-b_+^\dagger\tilde{\rho}-b_+^\dagger\tilde{\rho}b_-\Big)\Big(e^{-i(\Omega_k+\Omega_+)s}(\mathcal{N}(\Omega_k,T)+1)+e^{-i(\Omega_+-\Omega_k)s}\mathcal{N}(\Omega_k,T)\Big)\Big] \tag{157}$$

$$+\Big\{e^{-i(\Omega_+-\Omega_-)t}\Big[\Big(b_+b_-^\dagger\tilde{\rho}-b_-^\dagger\tilde{\rho}b_+\Big)\Big(e^{-i(\Omega_k+\Omega_-)s}(\mathcal{N}(\Omega_k,T)+1)+e^{-i(\Omega_--\Omega_k)s}\mathcal{N}(\Omega_k,T)\Big) \tag{158}$$

$$+\ \Big(b_-^\dagger b_+\tilde{\rho}-b_+\tilde{\rho}b_-^\dagger\Big)\Big(e^{-i(\Omega_k-\Omega_+)s}(\mathcal{N}(\Omega_k,T)+1)+e^{i(\Omega_++\Omega_k)s}\mathcal{N}(\Omega_k,T)\Big)\Big]\Big\} \tag{159}$$

After integrating we get:

$$-\frac{\sin\theta\cos\theta}{2\sqrt{\Omega_+\Omega_-}}\Big\{e^{i(\Omega_+-\Omega_-)t}\Big[\Big(b_+^\dagger b_-\tilde{\rho}-b_-\tilde{\rho}b_+^\dagger\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_-)\left(\mathcal{N}(\Omega_-,T)+1\right)+\Big(b_-b_+^\dagger\tilde{\rho}-b_+^\dagger\tilde{\rho}b_-\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_+)\mathcal{N}(\Omega_+,T)\Big] \tag{160}$$

$$+\Big\{e^{-i(\Omega_+-\Omega_-)t}\Big[\Big(b_+b_-^\dagger\tilde{\rho}-b_-^\dagger\tilde{\rho}b_+\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_-)\mathcal{N}(\Omega_-,T)+\Big(b_-^\dagger b_+\tilde{\rho}-b_+\tilde{\rho}b_-^\dagger\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_+)\left(\mathcal{N}(\Omega_+,T)+1\right)\Big]\Big\}, \tag{161}$$

which gives the following master equation:

$$\frac{\partial\tilde{\rho}}{\partial t}=\frac{\sin^2\theta}{\Omega_+}\mathcal{J}_{\mathrm{RC}}(\Omega_+)\Big[\mathcal{N}(\Omega_+,T)\Big(b_+^\dagger\tilde{\rho}b_+-\frac{1}{2}\Big\{b_+b_+^\dagger,\tilde{\rho}\Big\}\Big)+\left(\mathcal{N}(\Omega_+,T)+1\right)\Big(b_+\tilde{\rho}b_+^\dagger-\frac{1}{2}\Big\{b_+^\dagger b_+,\tilde{\rho}\Big\}\Big)\Big] \tag{162}$$

$$+\ \frac{\cos^2\theta}{\Omega_-}\mathcal{J}_{\mathrm{RC}}(\Omega_-)\Big[\mathcal{N}(\Omega_-,T)\Big(b_-^\dagger\tilde{\rho}b_--\frac{1}{2}\Big\{b_-b_-^\dagger,\tilde{\rho}\Big\}\Big)+\left(\mathcal{N}(\Omega_-,T)+1\right)\Big(b_-\tilde{\rho}b_-^\dagger-\frac{1}{2}\Big\{b_-^\dagger b_-,\tilde{\rho}\Big\}\Big)\Big] \tag{163}$$

$$-\ \frac{\sin\theta\cos\theta}{2\sqrt{\Omega_+\Omega_-}}\Big\{e^{i\Delta\Omega t}\Big[\Big(b_+^\dagger b_-\tilde{\rho}-b_-\tilde{\rho}b_+^\dagger\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_-)\left(\mathcal{N}(\Omega_-,T)+1\right)+\Big(b_-b_+^\dagger\tilde{\rho}-b_+^\dagger\tilde{\rho}b_-\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_+)\mathcal{N}(\Omega_+,T)\Big] \tag{164}$$

$$+\ e^{-i\Delta\Omega t}\Big[\Big(b_+b_-^\dagger\tilde{\rho}-b_-^\dagger\tilde{\rho}b_+\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_-)\mathcal{N}(\Omega_-,T)+\Big(b_-^\dagger b_+\tilde{\rho}-b_+\tilde{\rho}b_-^\dagger\Big)\mathcal{J}_{\mathrm{RC}}(\Omega_+)\left(\mathcal{N}(\Omega_+,T)+1\right)\Big]+\text{h.c.}\Big\}, \tag{165}$$

where we have defined $\Delta\Omega := \Omega_+ - \Omega_-$.

$$h(t)=\begin{pmatrix}\frac{\mathrm{s}_\theta^2}{\Omega_+}J_+\left(N_++1\right) & \frac{\mathrm{s}_\theta\,\mathrm{c}_\theta}{\sqrt{\Omega_+\Omega_-}}e^{-i(\Omega_+-\Omega_-)t}\left[J_+(N_++1)+J_-(N_-+1)\right] & 0 & 0\\ \frac{\mathrm{s}_\theta\,\mathrm{c}_\theta}{\sqrt{\Omega_+\Omega_-}}e^{+i(\Omega_+-\Omega_-)t}\left[J_+(N_++1)+J_-(N_-+1)\right] & \frac{\mathrm{c}_\theta^2}{\Omega_-}J_-\left(N_-+1\right) & 0 & 0\\ 0 & 0 & \frac{\mathrm{s}_\theta^2}{\Omega_+}J_+N_+ & \frac{\mathrm{s}_\theta\,\mathrm{c}_\theta}{\sqrt{\Omega_+\Omega_-}}e^{+i(\Omega_+-\Omega_-)t}\left[J_+N_++J_-N_-\right]\\ 0 & 0 & \frac{\mathrm{s}_\theta\,\mathrm{c}_\theta}{\sqrt{\Omega_+\Omega_-}}e^{-i(\Omega_+-\Omega_-)t}\left[J_+N_++J_-N_-\right] & \frac{\mathrm{c}_\theta^2}{\Omega_-}J_-N_-\end{pmatrix}$$

$$L_1 = \sqrt{\frac{\sin^2\theta}{\Omega_+}\mathcal{J}_{\mathrm{RC}}(\Omega_+)\mathcal{N}(\Omega_+,T)b_+^\dagger} \tag{166}$$

$$L_2 = \sqrt{\frac{\sin^2\theta}{\Omega_+}\mathcal{J}_{\mathrm{RC}}(\Omega_+)\left(\mathcal{N}(\Omega_+,T)+1\right)b_+} \tag{167}$$

$$L_3 = \sqrt{\frac{\cos^2\theta}{\Omega_-}\mathcal{J}_{\mathrm{RC}}(\Omega_-)\mathcal{N}(\Omega_-,T)b_-^\dagger} \tag{168}$$

$$L_4 = \sqrt{\frac{\cos^2\theta}{\Omega_-}\mathcal{J}_{\mathrm{RC}}(\Omega_-)\left(\mathcal{N}(\Omega_-,T)+1\right)b_-} \tag{169}$$

In the calculation we have neglected the terms containing $e^{\pm 2i\Omega_\pm t}$ because of the secular approximation, as those are the terms containing a subtraction of different system frequencies [?], which in our case are $\pm\Omega_\pm$, as they are the frequencies associated to the rotation of the operators into the interaction picture in Eq. (144).

From here, in fact we can build the matrix C, one just needs to multiply by the corresponding weights. We have

$$C_{\mathrm{Global}} = \begin{pmatrix} \frac{\sqrt{\mathcal{J}_{\mathrm{RC}}(\Omega_+)[\mathcal{N}(\Omega_+,T)+1)]}}{\sqrt{2\Omega_+}}\sin\theta \\ \frac{\sqrt{\mathcal{J}_{\mathrm{RC}}(\Omega_+)\mathcal{N}(\Omega_+,T)}}{\sqrt{2\Omega_+}}\sin\theta \\ \frac{\sqrt{\mathcal{J}_{\mathrm{RC}}(\Omega_-)[\mathcal{N}(\Omega_-,T)+1)]}}{\sqrt{2\Omega_-}}\cos\theta \\ \frac{\sqrt{\mathcal{J}_{\mathrm{RC}}(\Omega_-)\mathcal{N}(\Omega_-,T)}}{\sqrt{2\Omega_-}}\cos\theta \end{pmatrix} \bullet \begin{pmatrix} \sqrt{\Omega_+}\cos(\theta) & \frac{i\cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+}\sin(\theta) & \frac{-i\sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_+}\cos(\theta) & \frac{-i\cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+}\sin(\theta) & \frac{i\sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_-}\sin(\theta) & \frac{i\sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-}\cos(\theta) & \frac{i\cos(\theta)}{\sqrt{\Omega_-}} \\ \sqrt{\Omega_-}\sin(\theta) & \frac{-i\sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-}\cos(\theta) & \frac{-i\cos(\theta)}{\sqrt{\Omega_-}} \end{pmatrix}, \tag{170}$$

where $A \bullet B = \mathrm{diag}(A)B$ represents element-wise multiplication.

Regarding the Lamb shift terms, we just need to do the same procedure as in the local case to obtain the LS Hamiltonian:

$$H_{\mathrm{LS}} = \frac{\sin^2\theta}{\Omega_+}b_+^\dagger b_+ \mathrm{LS}^+ + \frac{\cos^2\theta}{\Omega_-}b_-^\dagger b_- \mathrm{LS}^-, \tag{171}$$

with

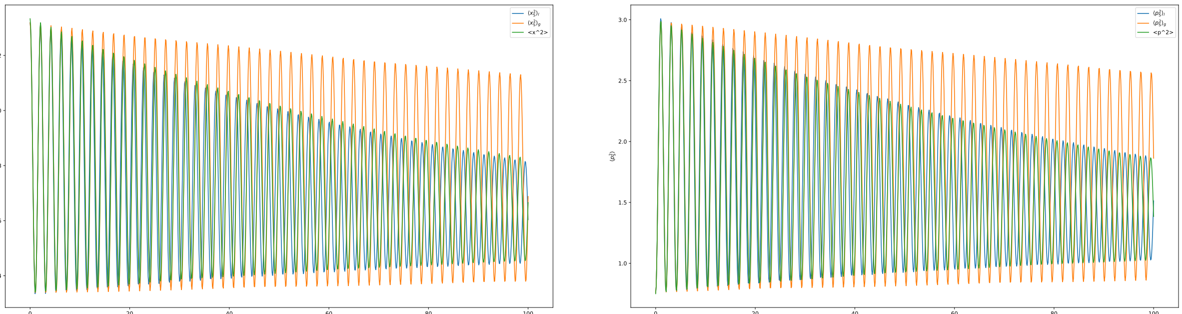
$$\mathrm{LS}^\pm = \frac{1}{4}\left(-\omega_m^2 + 2\Omega_\pm^2\right), \quad \omega_m^2 > \Omega_\pm^2. \tag{172}$$

To include the LS, we just need to add H_{LS} in G and compute A and D accordingly.

7.3 Thermalization

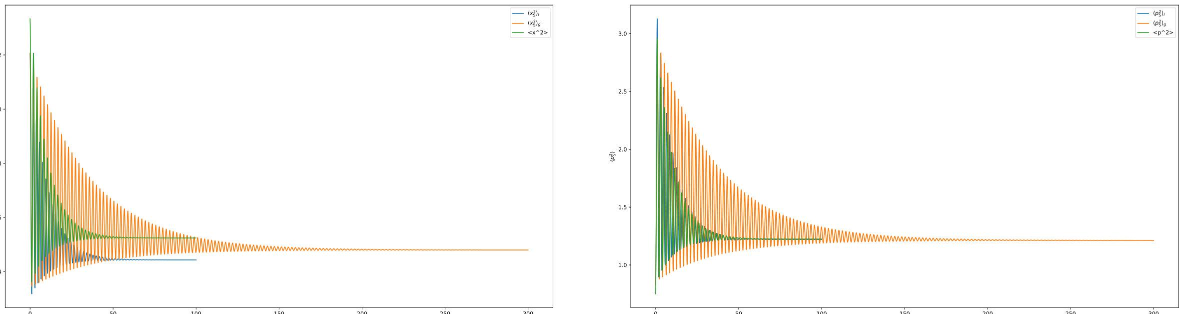
Let us now study the steady state by solving the Lyapunov equation $\frac{\mathrm{d}\sigma}{\mathrm{d}t} = 0$. To do that, we first compute the matrices A and D as usual. Let $c := \cos\theta$ and $s := \sin\theta$. The D matrix is then given by

$$D = \begin{pmatrix} \frac{\sin^2(2\theta)}{4\sigma_3}\left[J_-\Omega_+^2(2N_-+1) + J_+\Omega_-^2(2N_++1)\right] & 0 & \sigma_2 & 0 \\ 0 & \frac{\sin^2(2\theta)}{8}\left[J_-(2N_-+1) + J_+(2N_++1)\right] & 0 & \sigma_1 \\ \sigma_2 & 0 & \frac{J_-\Omega_+^2(2N_-+1)\cos^4\theta + J_+\Omega_-^2(2N_++1)\sin^4\theta}{\sigma_3} & 0 \\ 0 & \sigma_1 & 0 & \frac{1}{2}\left[J_-(2N_-+1)\cos^4\theta + J_+(2N_++1)\sin^4\theta\right] \end{pmatrix} \tag{173}$$



(a) $\left\langle x^2 \right\rangle(t)$ for the global ME. (b) $\left\langle p^2 \right\rangle(t)$ for the global ME.

Figure 3: $\omega_m = 10, \Omega_{\mathrm{S}} = 1.5, T = 1.0, \Gamma = 0.1$.



(a) $\left\langle x^2 \right\rangle(t)$ for the global ME. (b) $\left\langle p^2 \right\rangle(t)$ for the global ME.

Figure 4: $\omega_m = 10, \Omega_{\mathrm{S}} = 1.5, T = 1.0, \Gamma = 1.0$.

with

$$\sigma_1 = \frac{\sin(2\theta)}{4}\Big[J_-(2N_- + 1)\cos^2\theta - J_+(2N_+ + 1)\sin^2\theta\Big], \tag{174}$$

$$\sigma_2 = \frac{\sin(2\theta)}{2\,\sigma_3}\Big[J_-\Omega_+^2(2N_- + 1)\cos^2\theta - J_+\Omega_-^2(2N_+ + 1)\sin^2\theta\Big], \tag{175}$$

$$\sigma_3 = 2\,\Omega_+^2\,\Omega_-^2. \tag{176}$$

.

For the A matrix we obtain

$$A = \begin{pmatrix} -\frac{\sin^2(2\theta)}{8\,\Omega_+\Omega_-}\Big(J_-\Omega_+ + J_+\Omega_-\Big) & -1 & \sigma_1 & 0 \\ \Omega_s^2 & -\frac{\sin^2(2\theta)}{8\,\Omega_+\Omega_-}\Big(J_-\Omega_+ + J_+\Omega_-\Big) & -\lambda & \sigma_1 \\ \sigma_1 & 0 & \sigma_2 & -1 \\ -\lambda & \sigma_1 & \Omega_{\mathrm{rc}}^2 & \sigma_2 \end{pmatrix} \tag{177}$$

where

$$\sigma_1 = -\frac{\sin(2\theta)}{4\,\Omega_+\Omega_-}\Big[J_-\Omega_+\cos^2\theta + J_+\Omega_-(\cos^2\theta - 1)\Big] \tag{178}$$

$$= -\frac{\sin(2\theta)}{4\,\Omega_+\Omega_-}\Big(J_-\Omega_+ - J_+\Omega_-\Big)\cos^2\theta \;+\; \frac{\sin(2\theta)}{4\,\Omega_+\Omega_-}J_+\Omega_-, \tag{179}$$

$$e\sigma_2 = -\frac{1}{2\,\Omega_+\Omega_-}\Big[J_-\Omega_+(1 - 2\sin^2\theta + \sin^4\theta) + J_+\Omega_-\sin^4\theta\Big] \tag{180}$$

$$= -\frac{1}{2\,\Omega_+\Omega_-}\Big[J_-\Omega_+\cos^4\theta + J_+\Omega_-\sin^4\theta\Big]. \tag{181}$$

Now, it is a matter of solving the Lyapunov equation (numerically) and finding the equilibrium state.

can we say something at $\lambda \to 0$? as now the prefactors contain trigonometric functions, which also depend on λ!

7.4 Validity of the global master equation

In Ref. [**?**], the global GKLS master equation is said to be valid whenever $\min_{j\neq k}\{|\Omega_j - \Omega_k|, 2\Omega_j\} \gg \max_i\{\gamma_i\}$, where Ω_i is the frequency of the i-th normal mode and γ_i is the dissipation rate of the i-th bath. In our case, we just have two normal modes, so the validity condition will read $\min\{|\Omega_+ - \Omega_-|, 2\Omega_-\} \gg \max_i\{\gamma_i\}$.

7.5 Relation between spectral densities

The aim of this section is to show that the relation between the original spectral density and the RC spectral density is given by Eq. (127). We give an analogous proof to the one in the literature [?, ?, ?, ?], which is based on the fact that the Fourier space propagator in both coordinate systems must be the same. Starting with the Hamiltonian in Eq. (22), the Hamilton equations of motion lead to

$$\ddot{x}_{\mathrm{S}}(t) = -\Omega_{\mathrm{S}}^2 x_{\mathrm{S}}(t) - x_{\mathrm{S}}(t)\sum_k \frac{c_k^2}{\omega_k^2} + \sum_k x_k(t)c_k, \tag{182}$$

$$\ddot{x}_k(t) = -\omega_k^2 x_k(t) + c_k x_{\mathrm{S}}(t). \tag{183}$$

Let us now Fourier transform both equations (such that $\tilde{f}(z) = \int_{-\infty}^{+\infty} \mathrm{d}t\, f(z)e^{izt}$):

$$-z^2\tilde{x}_{\mathrm{S}}(t) = -\Omega_{\mathrm{S}}^2 \tilde{x}_{\mathrm{S}}(t) - \tilde{x}_{\mathrm{S}}(t)\sum_k \frac{c_k^2}{\omega_k^2} + \sum_k \tilde{x}_k(t)c_k, \tag{184}$$

$$-z^2\tilde{x}_k(t) = -\omega_k^2 \tilde{x}_k(t) + c_k \tilde{x}_{\mathrm{S}}(t). \tag{185}$$

Now, substituting the second equation into the first we get

$$-\Omega_{\mathrm{S}}^2 \tilde{x}_{\mathrm{S}}(t) = \tilde{\mathcal{L}}_{\mathrm{S}}(z)\tilde{x}_{\mathrm{S}}(t), \quad \tilde{\mathcal{L}}_{\mathrm{S}}(z) := -z^2 + \sum_k \frac{c_k^2}{\omega_k^2} - \sum_k \frac{c_k^2}{\omega_k^2 - z^2}. \tag{186}$$

Now, we have to do the same trick with the Hamiltonian in Eq. (134). In this case, we have 3 equations of motion, given by

$$\ddot{x}_{\mathrm{S}}(t) = -\Omega_{\mathrm{S}}^2 x_{\mathrm{S}}(t) - \delta\Omega_{\mathrm{S}}^2 x_{\mathrm{S}}(t) + \lambda X_{\mathrm{RC}}(t), \tag{187}$$

$$\ddot{X}_{\mathrm{RC}}(t) = -\Omega_{\mathrm{RC}}^2 X_{\mathrm{RC}}(t) + \lambda x_{\mathrm{S}}(t) + \sum_k C_k X_k(t), \tag{188}$$

$$\ddot{X}_k(t) = -\Omega_k^2 X_k(t) + C_k X_{\mathrm{RC}}(t). \tag{189}$$

Fourier-transforming we get:

$$-z^2\tilde{x}_{\mathrm{S}}(t) = -\Omega_{\mathrm{S}}^2 \tilde{x}_{\mathrm{S}}(t) - \delta\Omega_{\mathrm{S}}^2 \tilde{x}_{\mathrm{S}}(t) + \lambda \tilde{X}_{\mathrm{RC}}(t), \tag{190}$$

$$-z^2\tilde{X}_{\mathrm{RC}}(t) = -\Omega_{\mathrm{RC}}^2 \tilde{X}_{\mathrm{RC}}(t) + \lambda \tilde{x}_{\mathrm{S}}(t) + \sum_k C_k \tilde{X}_k(t), \tag{191}$$

$$-z^2\tilde{X}_k(t) = -\Omega_k^2 \tilde{X}_k(t) + C_k \tilde{X}_{\mathrm{RC}}(t), \tag{192}$$

from where we can obtain

$$-\Omega_{\mathrm{S}}^2 \tilde{x}_{\mathrm{S}} = \tilde{\mathcal{L}}(z)\tilde{x}_{\mathrm{S}}, \quad \tilde{\mathcal{L}}(z) := -z^2 + \delta\Omega_{\mathrm{S}}^2 - \frac{\lambda^2}{-z^2 + \Omega_{\mathrm{RC}}^2 - \sum_k \frac{C_k^2}{\Omega_k^2 - z^2}}. \tag{193}$$

As we said, both Fourier propagators must be equal, so the following relation must hold:

$$-z^2 + \sum_k \frac{c_k^2}{\omega_k^2} - \sum_k \frac{c_k^2}{\omega_k^2 - z^2} = -z^2 + \delta\Omega_{\mathrm{S}}^2 - \frac{\lambda^2}{-z^2 + \Omega_{\mathrm{RC}}^2 - \sum_k \frac{C_k^2}{\Omega_k^2 - z^2}} \Rightarrow W_0(z) = \frac{\lambda^2}{-z^2 + \Omega_{\mathrm{RC}}^2 - W_1(z)}, \tag{194}$$

where we have defined the Cauchy transform of $\mathcal{J}(\omega)$ as (same for RC)

$$W_0(z) = \frac{1}{\pi}\int_{-\infty}^{+\infty} \mathrm{d}\omega\, \frac{\mathcal{J}(\omega)}{\omega - z}, \tag{195}$$

which can be inverted to obtain

$$W_1(z) = \Omega_{\text{RC}}^2 - z^2 - \frac{\lambda^2}{W_0(z)}. \tag{196}$$

From here, Eq. (127) can be derived by taking the imaginary part of Eq. (196) and noticing that $J(\omega) = -\operatorname{Im} W_0^+(z)$.

Now, let's take one of the proposals for spectral densities given in Table I in Ref. [?]:

$$\mathcal{J}(\omega) = \Gamma \frac{\omega}{\omega_m}\sqrt{1 - \frac{\omega^2}{\omega_m^2}}\Theta(\omega_m - \omega). \tag{197}$$

We want to compute $\mathcal{J}_{\text{RC}}(\omega)$. To do that, we first compute λ^2 via Eq. (125) to obtain

$$\lambda^2 = \frac{\Gamma \omega_m^2}{8}. \tag{198}$$

Now, let us compute $W_0^+(\omega)$. To do that, we first compute $W_0(\omega + i\epsilon)$, which gives

$$W_0(\omega + i\epsilon) = \frac{\Gamma}{2}\left(1 - \frac{2}{1 + \sqrt{1 + \frac{\omega_m^2}{(\epsilon - i\omega)^2}}}\right). \tag{199}$$

Now, if we define $z := \omega + i\epsilon$, such that $\epsilon - i\omega = -iz$ we have:

$$W_0(z) = \frac{\Gamma}{2}\left(1 - \frac{2}{1 + \sqrt{1 - \frac{\omega_m^2}{z^2}}}\right) = \frac{\Gamma}{2}\left(1 - \frac{2}{1 + \frac{1}{z}\sqrt{z^2 - \omega_m^2}}\right) = \frac{\Gamma}{2(z + \sqrt{z^2 - \omega_m^2})}\left(\sqrt{z^2 - \omega_m^2} - z\right) \tag{200}$$

$$= -\frac{\Gamma}{2(z^2 - (z^2 - \omega_m^2))}(\sqrt{z^2 - \omega_m^2} - z)^2 = -\frac{\Gamma}{2\omega_m^2}(\sqrt{z^2 - \omega_m^2} - z)^2 = \frac{\Gamma}{2} + \frac{\Gamma z}{\omega_m^2}(\sqrt{z^2 - \omega_m^2} - z). \tag{201}$$

Now, using that $\lim_{\epsilon \to 0^+} \sqrt{z^2 - \omega_m^2} = i\sqrt{\omega_m^2 - \omega^2}$ we have

$$W_0^+ = \frac{\Gamma}{2} - \frac{\Gamma\omega^2}{\omega_m^2} + i\frac{\Gamma\omega}{\omega_m^2}\sqrt{\omega_m^2 - \omega^2} \Rightarrow \left|W_0^+\right|^2 = \frac{\Gamma^2}{4}, \tag{202}$$

and the RC spectral density is then given by

$$\mathcal{J}_{\text{RC}}(\omega) = \frac{\omega\omega_m}{2}\sqrt{1 - \frac{\omega^2}{\omega_m^2}}\Theta(\omega_m - \omega), \tag{203}$$

which keeps the cutoff! Just for completeness, the RC frequency is computed via Eq. (126) to give

$$\Omega_{\text{RC}} = \frac{\omega_m}{\sqrt{2}}. \tag{204}$$

Additionally, the frequency shifts are given by [Moha]Is ths l.h.s. to the power of two by any chance? (at least the top line)

$$\begin{aligned} \delta\Omega_S^2 &= \frac{\Gamma}{2}, \\ \delta\Omega_{RC} &= \frac{\omega_m}{2}. \end{aligned} \tag{205}$$

When we look for the decay rates, we should use $\mathcal{J}(\sqrt{\Omega_S^2 + \delta\Omega_S^2})$ and $\mathcal{J}_{RC}(\Omega_{RC})$.

Note that we should have $\Gamma \leq \sqrt{\omega_m^2 - \Omega_S^2}$, otherwise the spectral density at shifted frequency of the system and hence the decay rate of the system will vanish.

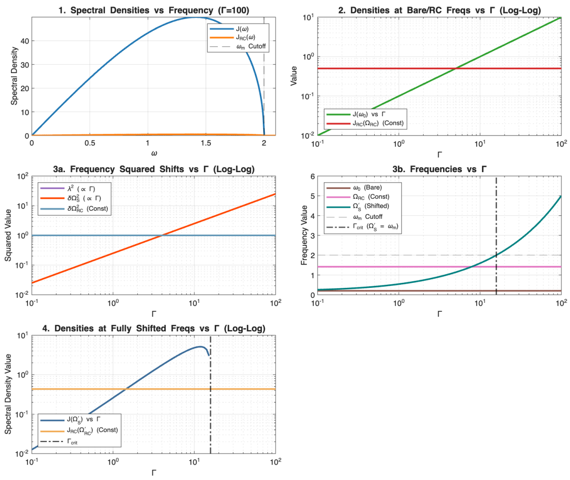


Figure 5: Caption

A (Local) GKLS master equation with reaction coordinate

Now, let us try to derive the (local) master equation (ME) for the system (harmonic oscillator) + RC. To do that, let us first identify some terms in the Hamiltonian as

$$H = \underbrace{\frac{1}{2}(p_S^2 + \Omega_S^2 x_S^2) + \frac{1}{2}\left[P_{\mathrm{RC}}^2 + \frac{\lambda^2}{\delta\Omega_{\mathrm{S}}^2}\left(X_{\mathrm{RC}} - \frac{\delta\Omega_{\mathrm{S}}^2}{\lambda}x_S\right)^2\right]}_{H_{\mathrm{S+RC}}} + \underbrace{\frac{1}{2}\sum_k \left(P_k^2 + \Omega_k^2 X_k^2\right)}_{H_{\mathrm{RB}}} \tag{206}$$

$$\underbrace{-\sum_k X_k C_k X_{\mathrm{RC}}}_{H_{\mathrm{I}}} + \underbrace{\frac{1}{2}\sum_k \frac{C_k^2}{\Omega_k^2} X_{\mathrm{RC}}^2}_{H_{\mathrm{S+RC}}},$$

so we can write the interacting Hamiltonian in the interaction picture as

$$\tilde{H}_{\mathrm{I}} = e^{i(H_{\mathrm{S+RC}}+H_{\mathrm{RB}})t} H_{\mathrm{I}} e^{-i(H_{\mathrm{S+RC}}+H_{\mathrm{RB}})t}. \tag{207}$$

Therefore, the dynamics of the system in the interaction picture reads: cite

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \underbrace{-i\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{I}}(t), \tilde{\rho}(0)\right]}_{(A)} \underbrace{- \int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{I}}(t), \left[\tilde{H}_{\mathrm{I}}(s), \tilde{\rho}(s)\right]\right]}_{(B)}. \tag{208}$$

Let us rewrite the interaction Hamiltonian as $H_{\mathrm{I}} = A \otimes B$, with $A := X_{\mathrm{RC}}$ and $B := -\sum_k C_k X_k$. Let us now treat each term of the ME separately:

$$(A) = -i\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{I}}(t), \tilde{\rho}(0)\right] = -i\,\mathrm{Tr}_B\left[\tilde{A}(t)\otimes\tilde{B}(t), \tilde{\rho}_{\mathrm{S+RC}}(0)\otimes\tilde{\rho}_B\right] = 0, \tag{209}$$

where we have used that $\mathrm{Tr}_B(\rho_B \tilde{B}(t)) = 0, \forall t$, where ρ_B is assumed to be in a Gibbs state. For the second term we have:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{I}}(t), \left[\tilde{H}_{\mathrm{I}}(s), \tilde{\rho}(s)\right]\right] = -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{I}}(t), \left[\tilde{H}_{\mathrm{I}}(t-s), \tilde{\rho}_{\mathrm{S+RC}}(t-s)\otimes\tilde{\rho}_B\right]\right] \tag{210}$$

$$= -\int_0^t \mathrm{d}s\,\mathrm{Tr}_B\left[\tilde{H}_{\mathrm{I}}(t)\tilde{H}_{\mathrm{I}}(t-s)\tilde{\rho}_{\mathrm{S+RC}}(t-s)\otimes\tilde{\rho}_B - \tilde{H}_{\mathrm{I}}(t-s)\tilde{\rho}_{\mathrm{S+RC}}(t-s)\otimes\tilde{\rho}_B\tilde{H}_{\mathrm{I}}(t) + \text{h.c.}\right], \tag{211}$$

where we changed $s \leftrightarrow t-s$. Now, we need to transform the operators into the interaction picture:

$$X_{\rm RC} = \frac{1}{\sqrt{2\Omega_{\rm RC}}}(a+a^\dagger) \Rightarrow \tilde{X}_{\rm RC} = \frac{1}{\sqrt{2\Omega_{\rm RC}}}(e^{-i\Omega_{\rm RC}t}a + e^{i\Omega_{\rm RC}t}a^\dagger), \tag{212}$$

$$B = -\frac{1}{\sqrt{2\Omega_k}}\sum_k C_k(b_k+b_k^\dagger) \Rightarrow \tilde{B} = -\frac{1}{\sqrt{2\Omega_k}}\sum_k C_k(e^{-i\Omega_k t}b_k + e^{i\Omega_k t}b_k^\dagger). \tag{213}$$

If we develop (B) in terms of $\tilde{A}$ and $\tilde{B}$ we get:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = -\int_0^t \mathrm{d}s\, \mathrm{Tr}_B\Big[(\tilde{A}(t)\otimes\tilde{B}(t))(\tilde{A}(t-s)\otimes\tilde{B}(t-s))\tilde{\rho}_{\rm S+RC}(t-s)\otimes\tilde{\rho}_B \tag{214}$$

$$-(\tilde{A}(t-s)\otimes\tilde{B}(t-s))\tilde{\rho}_{\rm S+RC}(t-s)\otimes\tilde{\rho}_B(\tilde{A}(t)\otimes\tilde{B}(t)) + \text{h.c.}\Big] \tag{215}$$

$$= -\int_0^t \mathrm{d}s\,\Big(\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_{\rm S+RC}(t-s) - \tilde{A}(t-s)\tilde{\rho}_{\rm S+RC}(t-s)\tilde{A}(t)\Big)\,\mathrm{Tr}_B\Big(\tilde{B}(t)\tilde{B}(t-s)\tilde{\rho}_B\Big) + \text{h.c.} \tag{216}$$

To simplify the expression, let us compute the bath-dependent term, which can be shown to be

$$\mathrm{Tr}_B\Big(\tilde{B}(t)\tilde{B}(t-s)\tilde{\rho}_B\Big) = \frac{1}{2}\sum_k \frac{C_k^2}{\Omega_k}\left[e^{-i\Omega_k s}\left(\mathcal{N}(\Omega_k,T)+1\right) + e^{i\Omega_k s}\mathcal{N}(\Omega_k,T)\right], \tag{217}$$

where $N(\Omega_k, T)$ is the bosonic occupation number of the mode with associated frequency Ω_k. Let us then simplify the expression:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = -\frac{1}{2}\sum_k \frac{C_k^2}{\Omega_k}\int_0^t \mathrm{d}s\,\Big(\tilde{A}(t)\tilde{A}(t-s)\tilde{\rho}_{\rm S+RC}(t) - \tilde{A}(t-s)\tilde{\rho}_{\rm S+RC}(t)\tilde{A}(t)\Big)\times \tag{218}$$

$$\times\left[e^{-i\Omega_k s}\left(\mathcal{N}(\Omega_k,T)+1\right) + e^{i\Omega_k s}\mathcal{N}(\Omega_k,T)\right] \tag{219}$$

$$= \frac{1}{4\Omega_{\rm RC}}\sum_k \frac{C_k^2}{\Omega_k}\int_0^\infty \mathrm{d}s\,\Big[e^{-i\Omega_{\rm RC}(2t-s)}a\tilde{\rho}_{\rm S+RC}(t)a + e^{i\Omega_{\rm RC}s}a\tilde{\rho}_{\rm S+RC}(t)a^\dagger + e^{-i\Omega_{\rm RC}s}a^\dagger\tilde{\rho}_{\rm S+RC}(t)a + \tag{220}$$

$$+\, e^{i\Omega_{\rm RC}(2t-s)}a^\dagger\tilde{\rho}_{\rm S+RC}(t)a^\dagger - \Big(e^{-i\Omega_{\rm RC}(2t-s)}a^2\tilde{\rho}_{\rm S+RC}(t) + e^{i\Omega_{\rm RC}s}a^\dagger a\tilde{\rho}_{\rm S+RC}(t) + e^{-i\Omega_{\rm RC}s}aa^\dagger\tilde{\rho}_{\rm S+RC}(t) + \tag{221}$$

$$+\, e^{i\Omega_{\rm RC}(2t-s)}(a^\dagger)^2\tilde{\rho}_{\rm S+RC}(t)\Big)\Big]\left[e^{-i\Omega_k s}\left(\mathcal{N}(\Omega_k,T)+1\right) + e^{i\Omega_k s}\mathcal{N}(\Omega_k,T)\right]. \tag{222}$$

If we expand the product and group the phases we get:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{4\Omega_{\rm RC}}\sum_k \frac{C_k^2}{\Omega_k}\int_0^\infty \mathrm{d}s \tag{223}$$

$$\Big[e^{-2i\Omega_{\rm RC}t}\left(a\tilde{\rho}_{\rm S+RC}(t)a - a^2\tilde{\rho}_{\rm S+RC}(t)\right)\Big(e^{-i(\Omega_k-\Omega_{\rm RC})s}(\mathcal{N}(\Omega_k,T)+1) + e^{i(\Omega_k+\Omega_{\rm RC})s}\mathcal{N}(\Omega_k,T)\Big) + \tag{224}$$

$$+\Big(a\tilde{\rho}_{\rm S+RC}(t)a^\dagger - a^\dagger a\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\Omega_k-\Omega_{\rm RC})s}(\mathcal{N}(\Omega_k,T)+1) + e^{i(\Omega_k+\Omega_{\rm RC})s}\mathcal{N}(\Omega_k,T)\Big) + \tag{225}$$

$$+\Big(a^\dagger\tilde{\rho}_{\rm S+RC}(t)a - aa^\dagger\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\Omega_k+\Omega_{\rm RC})s}(\mathcal{N}(\Omega_k,T)+1) + e^{i(\Omega_k-\Omega_{\rm RC})s}\mathcal{N}(\Omega_k,T)\Big) + \tag{226}$$

$$+\, e^{2i\Omega_{\rm RC}t}\Big(a^\dagger\tilde{\rho}_{\rm S+RC}(t)a^\dagger - (a^\dagger)^2\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\Omega_k+\Omega_{\rm RC})s}(\mathcal{N}(\Omega_k,T)+1) + e^{i(\Omega_k-\Omega_{\rm RC})s}\mathcal{N}(\Omega_k,T)\Big)\Big] \tag{227}$$

Now, because of the secular approximation, we can neglect the terms containing $e^{\pm 2i\Omega_{\rm RC}t}$, as those are the terms containing a subtraction of different system frequencies [?], which in our case are $\pm\Omega_{\rm RC}$, as they are the frequencies associated to the rotation of the operators into the interaction picture in Eq. (??). Then, the master equation reads

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{4\Omega_{\rm RC}}\sum_k \frac{C_k^2}{\Omega_k}\int_0^\infty \mathrm{d}s \tag{228}$$

$$\Big[\Big(a\tilde{\rho}_{\rm S+RC}(t)a^\dagger - a^\dagger a\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\Omega_k-\Omega_{\rm RC})s}(\mathcal{N}(\Omega_k,T)+1) + e^{i(\Omega_k+\Omega_{\rm RC})s}\mathcal{N}(\Omega_k,T)\Big) + \tag{229}$$

$$+\Big(a^\dagger\tilde{\rho}_{\rm S+RC}(t)a - aa^\dagger\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\Omega_k+\Omega_{\rm RC})s}(\mathcal{N}(\Omega_k,T)+1) + e^{i(\Omega_k-\Omega_{\rm RC})s}\mathcal{N}(\Omega_k,T)\Big)\Big] \tag{230}$$

Using that $\mathcal{J}_{\rm RC}(\omega) = \frac{\pi}{2}\sum_k \frac{C_k^2}{\Omega_k}\delta(\omega-\Omega_k)$ we can rewrite our expression as follows:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{2\pi\Omega_{\rm RC}}\int_0^\infty \mathrm{d}\omega\,\mathcal{J}_{\rm RC}(\omega)\int_0^\infty \mathrm{d}s \tag{231}$$

$$\Big[\Big(a\tilde{\rho}_{\rm S+RC}(t)a^\dagger - a^\dagger a\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\omega-\Omega_{\rm RC})s}(\mathcal{N}(\omega,T)+1) + e^{i(\omega+\Omega_{\rm RC})s}\mathcal{N}(\omega,T)\Big) + \tag{232}$$

$$+\Big(a^\dagger\tilde{\rho}_{\rm S+RC}(t)a - aa^\dagger\tilde{\rho}_{\rm S+RC}(t)\Big)\Big(e^{-i(\omega+\Omega_{\rm RC})s}(\mathcal{N}(\omega,T)+1) + e^{i(\omega-\Omega_{\rm RC})s}\mathcal{N}(\omega,T)\Big)\Big] \tag{233}$$

Now, using that

$$\int_0^\infty \mathrm{d}s\, e^{\pm i(\omega-\Omega_{\rm RC})s} = \pi\delta(\omega-\Omega_{\rm RC}) \pm i\mathbb{P}\frac{1}{\omega-\Omega_{\rm RC}} \tag{234}$$

$$\int_0^\infty \mathrm{d}s\, e^{\pm i(\omega+\Omega_{\rm RC})s} = \pi\delta(\omega+\Omega_{\rm RC}) \pm i\mathbb{P}\frac{1}{\omega+\Omega_{\rm RC}} \tag{235}$$

we can write:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{2\Omega_{\rm RC}}\Bigg[a\tilde{\rho}(t)a^\dagger\Bigg(\mathcal{J}_{\rm RC}(\Omega_{\rm RC})(\mathcal{N}(\Omega_{\rm RC},T)+1) - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)(\mathcal{N}+1)}{\omega-\Omega_{\rm RC}} + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)\mathcal{N}}{\omega+\Omega_{\rm RC}}\Bigg) + \tag{236}$$

$$+a^\dagger\tilde{\rho}(t)a\Bigg(\mathcal{J}_{\rm RC}(\Omega_{\rm RC})\mathcal{N}(\Omega_{\rm RC},T) + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)\mathcal{N}}{\omega-\Omega_{\rm RC}} - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)(\mathcal{N}+1)}{\omega+\Omega_{\rm RC}}\Bigg) + \tag{237}$$

$$-a^\dagger a\tilde{\rho}(t)\Bigg(\mathcal{J}_{\rm RC}(\Omega_{\rm RC})(\mathcal{N}(\Omega_{\rm RC},T)+1) - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)(\mathcal{N}+1)}{\omega-\Omega_{\rm RC}} + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)\mathcal{N}}{\omega+\Omega_{\rm RC}}\Bigg) + \tag{238}$$

$$-aa^\dagger\tilde{\rho}(t)\Bigg(\mathcal{J}_{\rm RC}(\Omega_{\rm RC})\mathcal{N}(\Omega_{\rm RC},T) + \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)\mathcal{N}}{\omega-\Omega_{\rm RC}} - \frac{i}{\pi}\mathbb{P}\int \mathrm{d}\omega\,\frac{\mathcal{J}_{\rm RC}(\omega)(\mathcal{N}+1)}{\omega+\Omega_{\rm RC}}\Bigg), \tag{239}$$

where we write $\tilde{\rho}(t) := \tilde{\rho}_{\rm S+RC}(t)$ and $\mathcal{N} := \mathcal{N}(\omega,T)$ to shorten the notation (we reincorporate all the labels later). When reincorporating the hermitian conjugate, we can simplify (B1) to a more familiar way:

$$\frac{\partial\tilde{\rho}(t)}{\partial t} = \frac{1}{\Omega_{\rm RC}}\mathcal{J}_{\rm RC}(\Omega_{\rm RC})\Big[\mathcal{N}(\Omega_{\rm RC},T)\Big(a^\dagger\tilde{\rho}a - \frac{1}{2}\Big\{aa^\dagger,\tilde{\rho}\Big\}\Big) + (\mathcal{N}(\Omega_{\rm RC},T)+1)\Big(a\tilde{\rho}a^\dagger - \frac{1}{2}\Big\{a^\dagger a,\tilde{\rho}\Big\}\Big)\Big] \tag{240}$$

$$+\,\text{Lamb shift terms} \tag{241}$$

To compute the Lamb shift terms, let us first define

$$\Delta_1^{\pm} = \frac{1}{2\pi}\mathbb{P}\int \mathrm{d}\omega\, \frac{\mathcal{J}_{\mathrm{RC}}(\omega)(\mathcal{N}(\omega,T)+1)}{\omega \pm \Omega_{\mathrm{RC}}}, \tag{242}$$

$$\Delta_2^{\pm} = \frac{1}{2\pi}\mathbb{P}\int \mathrm{d}\omega\, \frac{\mathcal{J}_{\mathrm{RC}}(\omega)\mathcal{N}(\omega,T)}{\omega \pm \Omega_{\mathrm{RC}}}. \tag{243}$$

It can then be shown that all the principal value terms reduce to **DOUBLE CHECK AGAIN!!**

$$\begin{aligned}
\text{Lamb shift terms} &= -\frac{i}{\Omega_{\mathrm{RC}}}\left\{\left[a^\dagger a, \tilde{\rho}_{\mathrm{S+RC}}(t)\right](\Delta_2^+ - \Delta_1^-) + \left[a a^\dagger, \tilde{\rho}_{\mathrm{S+RC}}(t)\right](\Delta_2^- - \Delta_1^+)\right\} && (244)\\
&= -\frac{i}{\Omega_{\mathrm{RC}}}\left[a^\dagger a, \tilde{\rho}_{\mathrm{S+RC}}(t)\right]\left\{(\Delta_2^+ + \Delta_2^-) - (\Delta_1^+ + \Delta_1^-)\right\} && (245)\\
&= -\frac{i}{\Omega_{\mathrm{RC}}}\left[a^\dagger a, \tilde{\rho}_{\mathrm{S+RC}}(t)\right]\mathrm{LS}, && (246)
\end{aligned}$$

where in the last line we used that $\left[a, a^\dagger\right] = 1$.

If we use the SD in Eq. (203), let us try to compute the Lamb shift terms:

$$\mathrm{LS} = (\Delta_2^+ + \Delta_2^-) - (\Delta_1^+ + \Delta_1^-) = -\frac{1}{2\pi}\mathbb{P}\int \mathrm{d}\omega\, \mathcal{J}_{\mathrm{RC}}(\omega)\left(\frac{1}{\omega - \Omega_{\mathrm{RC}}} + \frac{1}{\omega + \Omega_{\mathrm{RC}}}\right) \tag{247}$$

$$= -\frac{1}{2\pi}\mathbb{P}\int \mathrm{d}\omega\, \mathcal{J}_{\mathrm{RC}}(\omega)\frac{2\omega}{\omega^2 - \Omega_{\mathrm{RC}}^2} = -\frac{\omega_m}{2\pi}\mathbb{P}\int \mathrm{d}\omega \sqrt{1 - \frac{\omega^2}{\omega_m^2}}\,\Theta(\omega_m - \omega)\,\frac{\omega^2}{\omega^2 - \Omega_{\mathrm{RC}}^2}, \tag{248}$$

which gives

$$\mathrm{LS} = \begin{cases} \frac{1}{4}\left(-\omega_m^2 + 2\Omega_{\mathrm{RC}}^2\right) & \text{if } \Omega_{\mathrm{RC}} < \omega_m \\ \frac{1}{4}\left(-\omega_m^2 + 2\Omega_{\mathrm{RC}}\left(\Omega_{\mathrm{RC}} - \sqrt{\Omega_{\mathrm{RC}}^2 - \omega_m^2}\right)\right) & \text{if } \Omega_{\mathrm{RC}} \geq \omega_m \end{cases} \tag{249}$$

Note that Eq. (249) vanishes in the case $\Omega_{\mathrm{RC}} < \omega_m$ because of Eq. (126), which gives $\Omega_{\mathrm{RC}} = \omega_m/\sqrt{2}$.

A.1 Numerical solution

We have showed that the Lindbladian master equation for a single RC when dealing with an harmonic oscillator Hamiltonian is given by

$$\frac{\partial\tilde{\rho}}{\partial t} = \frac{1}{\Omega_{\mathrm{RC}}}\mathcal{J}_{\mathrm{RC}}(\Omega_{\mathrm{RC}})\left[\mathcal{N}(\Omega_{\mathrm{RC}},T)\left(a^\dagger\tilde{\rho}a - \frac{1}{2}\left\{aa^\dagger,\tilde{\rho}\right\}\right) + \left(\mathcal{N}(\Omega_{\mathrm{RC}},T)+1\right)\left(a\tilde{\rho}a^\dagger - \frac{1}{2}\left\{a^\dagger a,\tilde{\rho}\right\}\right)\right], \tag{250}$$

as the Lamb shift term vanishes if $\Omega_{\mathrm{RC}} < \omega_m$, which is a physically-meaning regime as the cutoff ω_m can be, in general, very large.

In parallel, it is known that for a Lindbladian master equation of the form

$$\frac{\partial\tilde{\rho}}{\partial t} = \sum_k\left(L_k\tilde{\rho}L_k^\dagger - \frac{1}{2}\left\{L_k^\dagger L_k,\tilde{\rho}\right\}\right) \tag{251}$$

the equations of motion for the first and second moments are given by

$$\frac{\mathrm{d}\langle R\rangle}{\mathrm{d}t} = A\langle R\rangle, \tag{252}$$

$$\frac{\mathrm{d}\sigma}{\mathrm{d}t} = A\sigma + \sigma A^{\mathrm{T}} + D, \tag{253}$$

with $R := \left(x_{\mathrm{S}}, x_{\mathrm{RC}}; p_{\mathrm{S}}, p_{\mathrm{RC}}\right)^T$ being the vector of quadratures. Moreover, the matrices A and D are defined as follows:

$$A := -i\Omega\left[G - \mathrm{Im}(CC^\dagger)\right] \tag{254}$$

$$D := \Omega\,\mathrm{Re}(CC^\dagger)\Omega \tag{255}$$

with

$$\boldsymbol{\Omega} := i\hbar\begin{pmatrix} 0 & \mathbb{I}_2 \\ -\mathbb{I}_2 & 0 \end{pmatrix}, \quad G := \begin{pmatrix} \Omega_{\mathrm{S}}^2 & -\lambda & 0 & 0 \\ -\lambda & \Omega_{\mathrm{RC}}^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C := (c_1^{\mathrm{T}}; c_2^{\mathrm{T}})^{\mathrm{T}}, \tag{256}$$

[Moha]probably should be changed to this [?]

$$A := -i\Omega\left[G + \mathrm{Im}(C^\dagger C)\right] \tag{257}$$

$$D := \Omega\,\mathrm{Re}(C^\dagger C)\Omega \tag{258}$$

with

$$\boldsymbol{\Omega} := i\hbar\begin{pmatrix} 0 & \mathbb{I}_2 \\ -\mathbb{I}_2 & 0 \end{pmatrix}, \quad G := \begin{pmatrix} \Omega_{\mathrm{S}}^2 & -\lambda & 0 & 0 \\ -\lambda & \Omega_{\mathrm{RC}}^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C := (c_1, c_2)^{\mathrm{T}}, \tag{259}$$

where c_k is defined such that the Lindblad operators are given by $L_k = c_k^{\mathrm{T}} R$. For our master equation, the Lindblad operators can be shown to be

$$L_1 = \sqrt{\kappa N}\, a_{\mathrm{RC}}^\dagger, \tag{260}$$

$$L_2 = \sqrt{\kappa(N+1)}\, a_{\mathrm{RC}}, \tag{261}$$

with $\kappa := \frac{\mathcal{J}(\Omega_{\mathrm{RC}})}{\Omega_{\mathrm{RC}}}$ and $N := \mathcal{N}(\Omega_{\mathrm{RC}}, T)$. Therefore, the vectors c_k, with $k \in \{1, 2\}$, are given by

$$c_1 = \sqrt{\frac{\kappa N}{2\hbar}}\begin{pmatrix} 0 \\ \sqrt{\Omega_{\mathrm{RC}}} \\ 0 \\ -\,i/\sqrt{\Omega_{\mathrm{RC}}} \end{pmatrix}, \qquad c_2 = \sqrt{\frac{\kappa(N+1)}{2\hbar}}\begin{pmatrix} 0 \\ \sqrt{\Omega_{\mathrm{RC}}} \\ 0 \\ i/\sqrt{\Omega_{\mathrm{RC}}} \end{pmatrix}. \tag{262}$$

With all these definitions, we can now compute A and D. Let us start by explicitly writing down C:

$$C = \begin{pmatrix} 0 & 0 \\ \sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{RC}})N}{2}} & \sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{RC}})(N+1)}{2}} \\ 0 & 0 \\ -\frac{i}{\Omega_{\mathrm{RC}}}\sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{RC}})N}{2}} & \frac{i}{\Omega_{\mathrm{RC}}}\sqrt{\frac{\mathcal{J}(\Omega_{\mathrm{RC}})(N+1)}{2}} \end{pmatrix}. \tag{263}$$

Now, let us compute $CC^\dagger$:

$$CC^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \dfrac{\Omega_{\mathrm{RC}}\,\kappa(2N+1)}{2\hbar} & 0 & -\dfrac{i\kappa}{2\hbar} \\ 0 & 0 & 0 & 0 \\ 0 & \dfrac{i\kappa}{2\hbar} & 0 & \dfrac{\kappa(2N+1)}{2\hbar\,\Omega_{\mathrm{RC}}} \end{pmatrix}. \tag{264}$$

Finally, applying the definitions we get:

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & -\frac{1}{2}\frac{\mathcal{J}(\Omega_{\mathrm{RC}})}{\Omega_{\mathrm{RC}}} & 0 & 1 \\ -\Omega_{\mathrm{S}}^2 & \lambda & 0 & 0 \\ \lambda & -\Omega_{\mathrm{RC}}^2 & 0 & -\frac{1}{2}\frac{\mathcal{J}(\Omega_{\mathrm{RC}})}{\Omega_{\mathrm{RC}}} \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\mathcal{J}(\Omega_{\mathrm{RC}})}{2\,\Omega_{\mathrm{RC}}}(2N+1) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \Omega_{\mathrm{RC}}\frac{\mathcal{J}(\Omega_{\mathrm{RC}})}{2}(2N+1) \end{pmatrix}. \tag{265}$$

Now, solving the Lyapunov equation for the second moments (that is, taking $\frac{\mathrm{d}\sigma}{\mathrm{d}t} = 0$) we can obtain the quadratures at equilibrium, both of the system and the RC. If we do that, we obtain the following result:

$$\sigma_{\mathrm{eq}} = \begin{pmatrix} \frac{(2N+1)\left(\kappa^4 + 4\kappa^2\Omega_{\mathrm{S}}^2 + 8\kappa^2\Omega_{\mathrm{RC}}^2 - 16\lambda^2 + 16\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2 + 16\Omega_{\mathrm{RC}}^4\right)}{16\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} & \frac{\lambda(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{RC}}^2\right)}{4\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} & 0 & \frac{\kappa\lambda(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{RC}}^2\right)}{8\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} \\ \frac{\lambda(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{RC}}^2\right)}{4\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} & \frac{(2N+1)\left(\kappa^2\Omega_{\mathrm{S}}^2 - 2\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)}{2\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} & 0 & \frac{\kappa\lambda^2(2N+1)}{2\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} \\ 0 & 0 & \frac{(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{S}}^2 + 4\Omega_{\mathrm{RC}}^2\right)}{16\Omega_{\mathrm{RC}}} & -\frac{\lambda(2N+1)}{4\Omega_{\mathrm{RC}}} \\ \frac{\kappa\lambda(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{RC}}^2\right)}{8\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} & \frac{\kappa\lambda^2(2N+1)}{2\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} & -\frac{\lambda(2N+1)}{4\Omega_{\mathrm{RC}}} & \frac{(2N+1)\left(\kappa^2\lambda^2 + 2\kappa^2\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2 - 8\lambda^2\Omega_{\mathrm{RC}}^2 + 8\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^4\right)}{4\Omega_{\mathrm{RC}}\left(\kappa^2\Omega_{\mathrm{S}}^2 - 4\lambda^2 + 4\Omega_{\mathrm{S}}^2\Omega_{\mathrm{RC}}^2\right)} \end{pmatrix}, \tag{266}$$

which in the limit of weak coupling ($\lambda \to 0$) reads

$$\lim_{\lambda\to 0} \sigma_{\mathrm{eq}} = \begin{pmatrix} \frac{(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{S}}^2 + 4\Omega_{\mathrm{RC}}^2\right)}{16\,\Omega_{\mathrm{S}}^2\,\Omega_{\mathrm{RC}}} & 0 & 0 & 0 \\ 0 & \frac{2N+1}{2\Omega_{\mathrm{RC}}} & 0 & 0 \\ 0 & 0 & \frac{(2N+1)\left(\kappa^2 + 4\Omega_{\mathrm{S}}^2 + 4\Omega_{\mathrm{RC}}^2\right)}{16\,\Omega_{\mathrm{RC}}} & 0 \\ 0 & 0 & 0 & \frac{\Omega_{\mathrm{RC}}(2N+1)}{2} \end{pmatrix}. \tag{267}$$

Therefore, we observe that the system quadratures do not thermalize to the usual $N+1/2$ state, as the RC does. This can be understood from the fact that the Lindblad operators are local, therefore just coupling the RC to the bath, but not the system. Also, note that if the coupling is not weak, the RC also thermalizes to a different state than the $N+1/2$. Now, let us focus on the global master equation approach, where the normal modes will be linear combinations of operators of both bath and system, therefore allowing the system to also interact with the bath (via the whole S+RC composite).

B Ohmic spectral density

$$\mathcal{J}_{\mathrm{SB}}(\omega) = \frac{\alpha\omega\omega_c}{\omega^2 + \omega_c^2} \tag{268}$$

$$\mathcal{J}_{\mathrm{RC}}(\omega) = \gamma\omega e^{-\omega/\Lambda} \tag{269}$$

$$\alpha = \frac{2\lambda^2}{\pi\Omega_{\mathrm{RC}}}, \quad \omega_c = \frac{\Omega_{\mathrm{RC}}}{2\pi\gamma} \tag{270}$$

C A second RC

$$\mathcal{J}_{\mathrm{RC}}^{(1)} = \frac{\omega\omega_m}{2}\sqrt{1 - \frac{\omega^2}{\omega_m^2}}\,\Theta(\omega_m - \omega), \quad \Omega_{\mathrm{RC}}^{(1)} = \frac{\omega_m}{\sqrt{2}}. \tag{271}$$

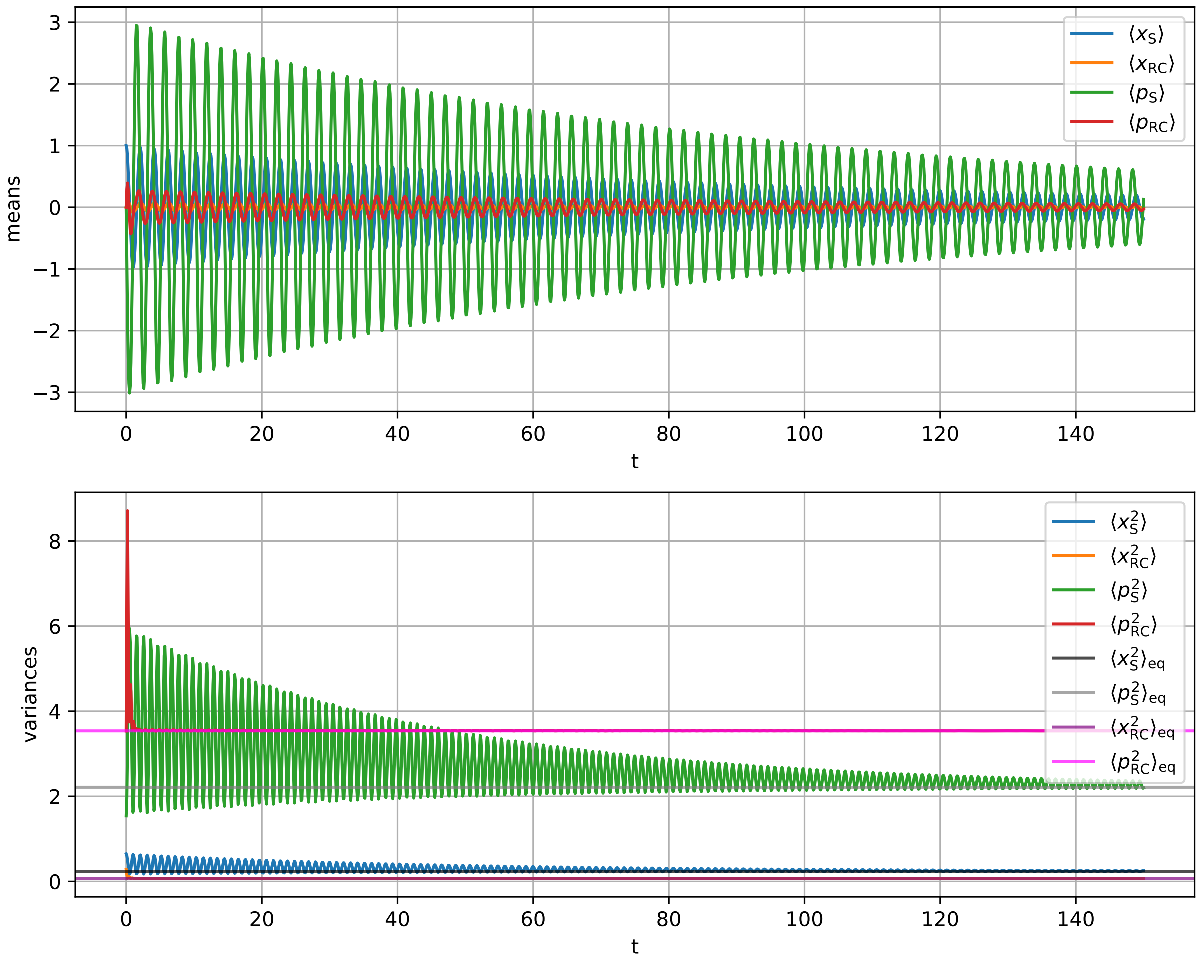


Figure 6: $\Gamma = 1.0, \Omega_S = 3.0, \omega_m = 10.0$

The Hamiltonian in this case will be given by

$$H^{2-\mathrm{RC}} = \frac{1}{2}\left[P_{\mathrm{S}}^2 + (\Omega_{\mathrm{S}}^2 + \delta\Omega_{\mathrm{S}}^2)X_{\mathrm{S}}^2\right] + \frac{1}{2}\left(P_{\mathrm{RC}_1}^2 + \Omega_{\mathrm{RC}}^2 X_{\mathrm{RC}_1}^2\right) - \lambda^{(1)} X_{\mathrm{S}} X_{\mathrm{RC}_1} \tag{272}$$

$$+ \frac{1}{2}\left[P_{\mathrm{RC}_2}^2 + \Omega_{\mathrm{RC}}^2 X_{\mathrm{RC}_2}^2\right] + \frac{1}{2}\sum_k \left(P_k^2 + \Omega_k'^2 X_k^2\right) - \lambda^{(2)} X_{\mathrm{RC}_2} \sum_k C_k' X_k. \tag{273}$$

Using the same expressions as before we can compute the different quantities for the second RC, obtaining

$$\left|W_0^+\right|^2 = \frac{\omega_m^4}{16} \tag{274}$$

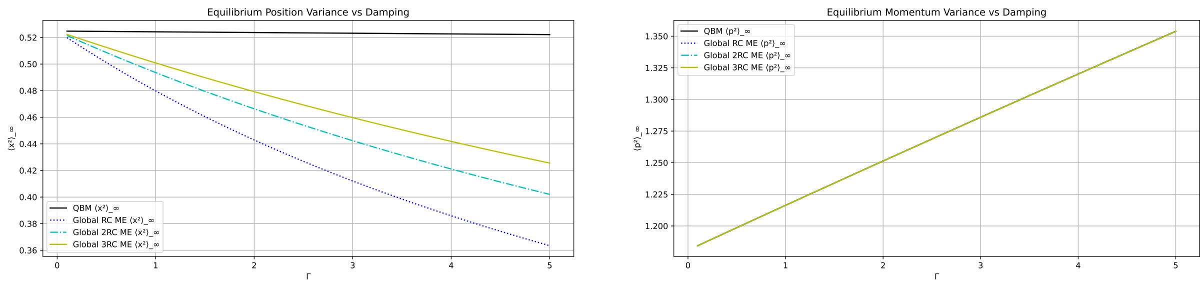
$$\lambda^{(2)^2} = \frac{\omega_m^4}{16} \tag{275}$$

$$\Omega_{\mathrm{RC}}^{(2)} = \frac{\omega_m^2}{2} = \Omega_{\mathrm{RC}}^{(1)} \quad \textbf{!!!!} \tag{276}$$

$$\mathcal{J}_{\mathrm{RC}}^{(2)} = \mathcal{J}_{\mathrm{RC}}^{(1)} \quad \textbf{!!!!} \tag{277}$$

Note it's like defining $\Gamma := \omega_m^2/2$ and all the quantities are equivalent, so applying again the mapping does not change neither the SD nor the couplings!

Therefore, extracting more RCs will help us gain more information about the bath, and the procedure is analogous at each round, it's just a matter of enlarging the dimensions of the matrices that enter the calculations.



(a) Equilibrium value of $\langle x^2\rangle$ for different values of Γ.

(b) Equilibrium value of $\langle p^2\rangle$ for different values of Γ.

Figure 7: $\omega_m = 5, \Omega_{\mathrm{S}} = 1.5, T = 1.0$

D Other stuff

D.1 From Lorentz-Drude to ohmic

Now, suppose that the original spectral density is given by

$$\mathcal{J}(\omega) = \frac{\alpha\omega_c\omega}{\omega^2 + \omega_c^2}. \tag{278}$$

To obtain the RC spectral density, we just need to compute $W_0^+(\omega)$:

$$W_0^+(\omega) = \lim_{\epsilon\searrow 0}\left(\frac{\alpha\omega_c}{\pi}\int_{-\infty}^{+\infty} \mathrm{d}\omega' \, \frac{\frac{\omega'}{\omega'^2+\omega_c^2}}{\omega' - (\omega + i\epsilon)}\right) = \frac{\alpha\omega_c}{\omega_c - i\omega}, \tag{279}$$

which gives

$$\mathcal{J}_{\mathrm{RC}}(\omega) = \lambda^2\left(\frac{\alpha\omega_c\omega}{\omega^2 + \omega_c^2}\right)\frac{|\omega_c - i\omega|^2}{\alpha^2\omega_c^2} = \frac{\lambda^2}{\alpha\omega_c}\omega = \eta\omega, \tag{280}$$

which is ohmic. Other Ahsan nazir reference [?]
New RC paper [?]

References